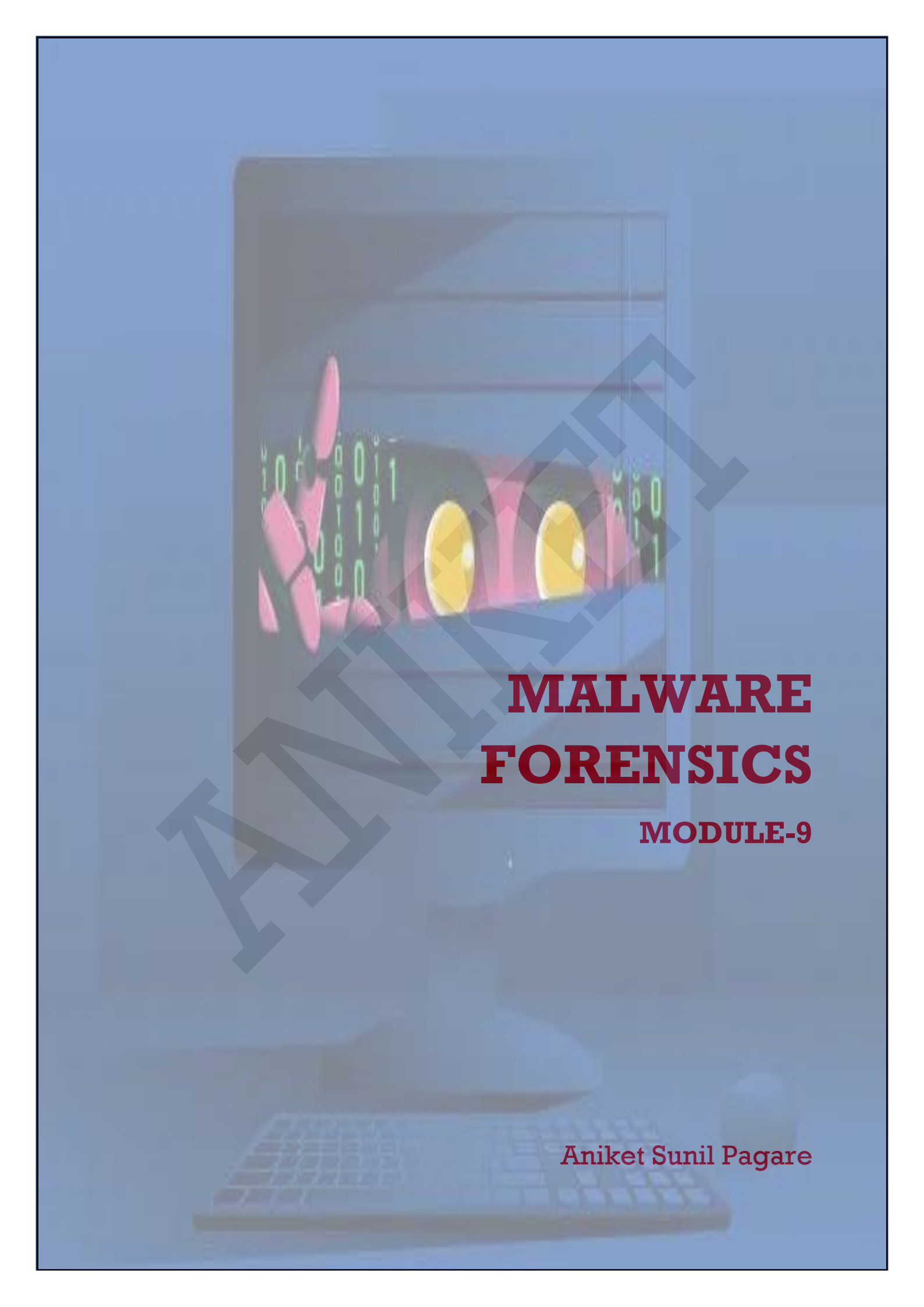




MALWARE FORENSICS

MODULE-9

Aniket Sunil Pagare

Malware Forensics – Table of Contents

1. Introduction

2. Objectives of Malware Forensics

3. Differences: Malware Analysis vs Malware Forensics

4. Types of Evidence in Malware Forensics

5. Malware Forensics Investigation Process

6. Common Malware Artifacts

7. Anti-Forensics in Malware

8. Malware Forensics Tools

9. Indicators of Compromise (IOCs)

10. Hands-on Labs

Lab 1: Perform Static Analysis on a Suspicious File

- **A. Using PEiD**
- **B. Using PEStudio**
- **C. Using IDA Pro**

Lab 2: Perform Static Analysis on a Suspicious File Using VirusTotal

Lab 3: Forensics Examination of a Suspicious PDF File

- **A. Using Didier Stevens Suite**
- **B. Using PDF Stream Dumper**

Lab 4: Examine a Suspicious File Using Online Resources

Lab 5: Examine Windows Event Logs

Malware Forensics –

1. Introduction

Malware forensics is a specialized branch of **digital forensics** that focuses on:

- **Identifying** malware on compromised systems.
- **Preserving** and **acquiring** malware evidence without contamination.
- **Analyzing** malware artifacts to determine infection origin, scope, and damage.
- **Supporting legal action** by maintaining proper evidence handling.

It is used in:

- **Incident Response**
- **Law Enforcement Investigations**
- **Threat Hunting**
- **Security Research**

2. Objectives of Malware Forensics

1. **Confirm infection** and identify malware type.
 2. **Preserve volatile and non-volatile evidence.**
 3. **Analyze infection vector** (email, USB, web exploit).
 4. **Determine malware behavior** (data theft, persistence, lateral movement).
 5. **Identify Indicators of Compromise (IOCs).**
 6. **Map the timeline** of the attack.
 7. **Attribute attack** if possible (threat actor, campaign).
 8. **Support remediation & prevention.**
-

3. Differences: Malware Analysis vs Malware Forensics

Aspect	Malware Analysis	Malware Forensics
Goal	Understand malware behavior & code	Investigate malware incident for evidence
Focus	Technical behavior	Legal & investigative
Tools	Sandboxes, debuggers	Forensic imaging, chain of custody
Output	IOCs, detection rules	Legal evidence, incident report

4. Types of Evidence in Malware Forensics

1. Volatile Data (RAM)

- Running processes
- Open network connections
- Decrypted payloads
- Malware in memory (fileless malware)
- Tools: **Volatility**, **Rekall**, **MemProcFS**

2. Non-Volatile Data

- Malware files on disk
- Registry keys
- Log files (Windows Event Logs, syslog)
- Scheduled tasks
- Browser artifacts

3. Network Data

- PCAPs of malicious traffic
- C2 server connections
- DNS lookups

5. Malware Forensics Investigation Process

Step 1 – Incident Identification

- Signs of infection:
 - Unusual network activity
 - Unknown processes
 - File modifications
 - User complaints (slow system, pop-ups)

Step 2 – Evidence Preservation

- Create **bit-by-bit image** of affected system:

`dd if=/dev/sda of=image.dd bs=4M conv=noerror,sync`

- Acquire memory dump (Windows: **DumpIt**, Linux: **LiME**).
- **Chain of custody**:
 - Document who collected the evidence, when, and how.
 - Store hashes (MD5/SHA256) for integrity.

Step 3 – Malware Identification

- Hash matching against:
 - VirusTotal
 - Hybrid Analysis
 - Threat intelligence feeds
- YARA rules scanning

Step 4 – Static Examination

- File metadata
- PE/ELF header analysis
- String extraction
- Detect packing/obfuscation

Step 5 – Dynamic Examination

- Execute in sandbox (Cuckoo, Any.Run, Joe Sandbox)

- Observe:
 - File system changes
 - Registry changes
 - Network traffic

Step 6 – Memory Analysis

- Use Volatility plugins:

`vol.py -f memdump.raw windows.pslist`

`vol.py -f memdump.raw malfind`

`vol.py -f memdump.raw netscan`

- Find injected processes, decrypted data.

Step 7 – Log Analysis

- Windows Event Viewer logs
- Syslog (Linux)
- Firewall & proxy logs

Step 8 – Timeline Creation

- Reconstruct attack sequence using:
 - Autopsy
 - Plaso/log2timeline

Step 9 – Reporting

- Document findings:
 - Malware name/type
 - IOCs
 - Entry point
 - Spread method
 - Impact

6. Common Malware Artifacts

On Windows

- %APPDATA%, %TEMP% suspicious files
- Registry run keys:

HKCU\Software\Microsoft\Windows\CurrentVersion\Run

- Scheduled tasks
- Event logs:
 - Security (ID 4624, 4625)
 - System
 - Application

On Linux

- /tmp hidden binaries
- /etc/cron* persistence
- .bash_history anomalies
- Suspicious processes:

ps aux | grep <name>

7. Anti-Forensics in Malware

Attackers often:

- **Wipe logs**
 - **Encrypt payloads**
 - **Pack binaries**
 - **Use fileless execution**
 - **Timestamp modification** (Timestomping)
-

8. Malware Forensics Tools

Disk Forensics

- Autopsy / Sleuth Kit
- FTK Imager
- EnCase

Memory Forensics

- Volatility
- Rekall
- MemProcFS

Network Forensics

- Wireshark
- Zeek
- NetworkMiner

Malware Analysis

- IDA Pro, Ghidra
- Detect It Easy (DIE)
- Cuckoo Sandbox
- Any.Run

Threat Intelligence

- VirusTotal
- MISP
- MalwareBazaar

9. Indicators of Compromise (IOCs)

- File hashes (MD5/SHA1/SHA256)
- IP addresses/domains
- Registry keys

- Mutex names
 - YARA rule matches
 - Filenames
-

ANTIKEY

Lab-1 Perform Static Analysis On A Suspicious file

Perform Static Analysis on a Suspicious File" means examining a potentially malicious file **without running it**, to gather information about its structure, code, and indicators of compromise (IOCs).

It's the **first step** in malware analysis and malware forensics, done to **minimize risk** and **quickly understand** what the file might do.

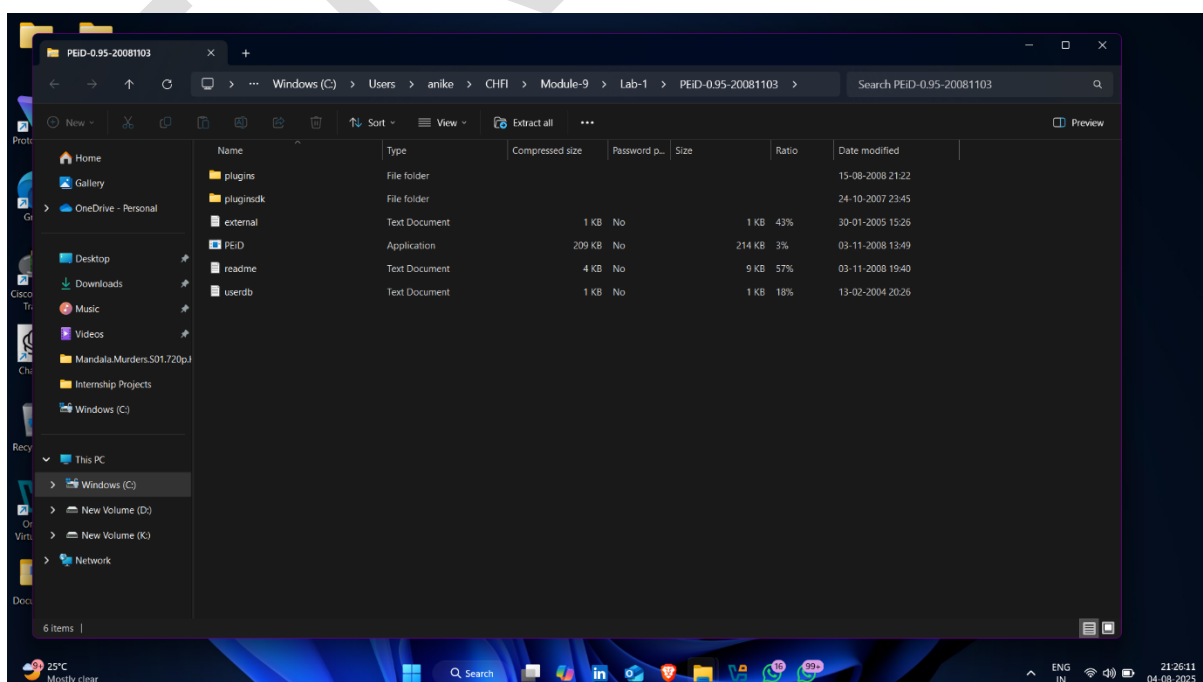
A) Perform Static Analysis On A Suspicious File Using PEid

PEiD (Portable Executable Identifier) is a **Windows-based malware analysis tool** used mainly for **detecting packers, cryptors, and compilers** in PE (Portable Executable) files such as .exe and .dll.

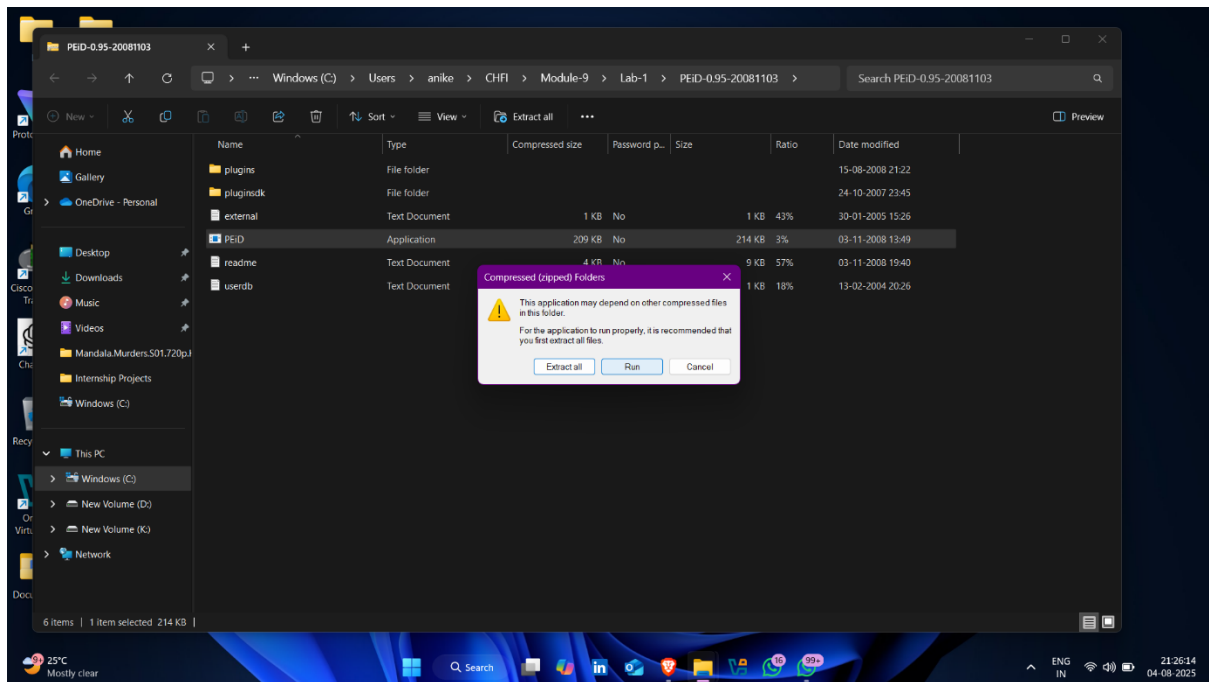
It is widely used in **malware forensics** and **reverse engineering** during **static analysis** to determine if a suspicious file is **packed or obfuscated**.

How to use it -:

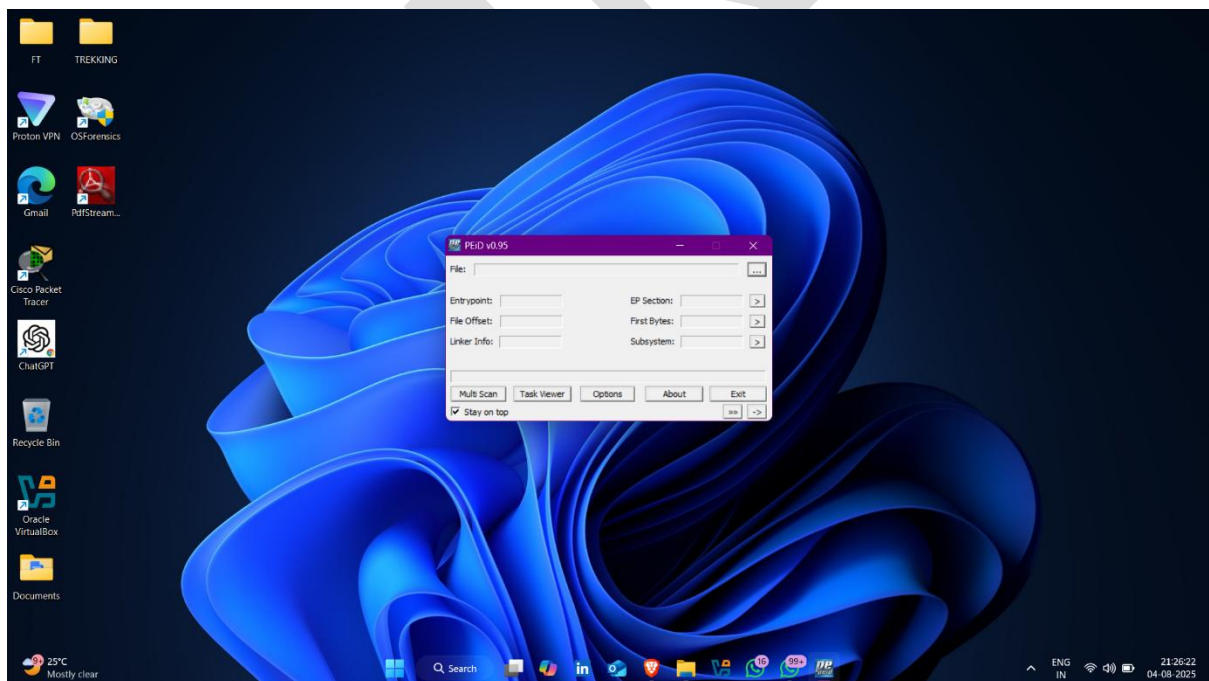
- Click on **PEid.exe** file



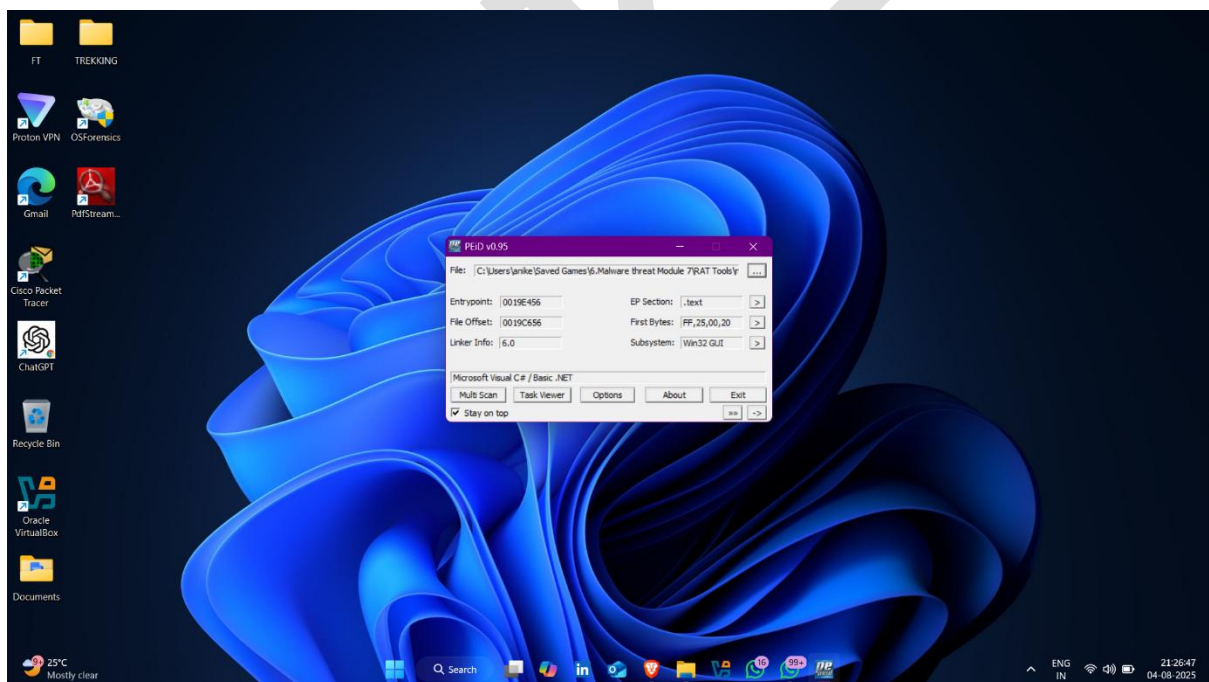
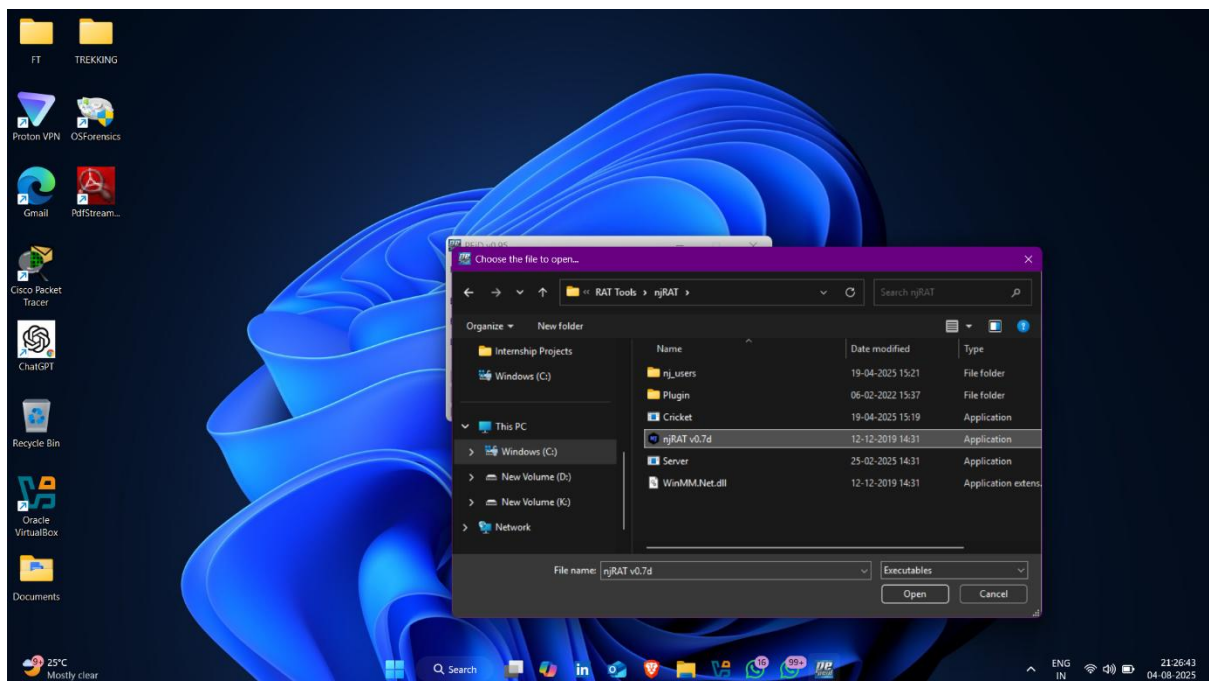
- Click on **Run**

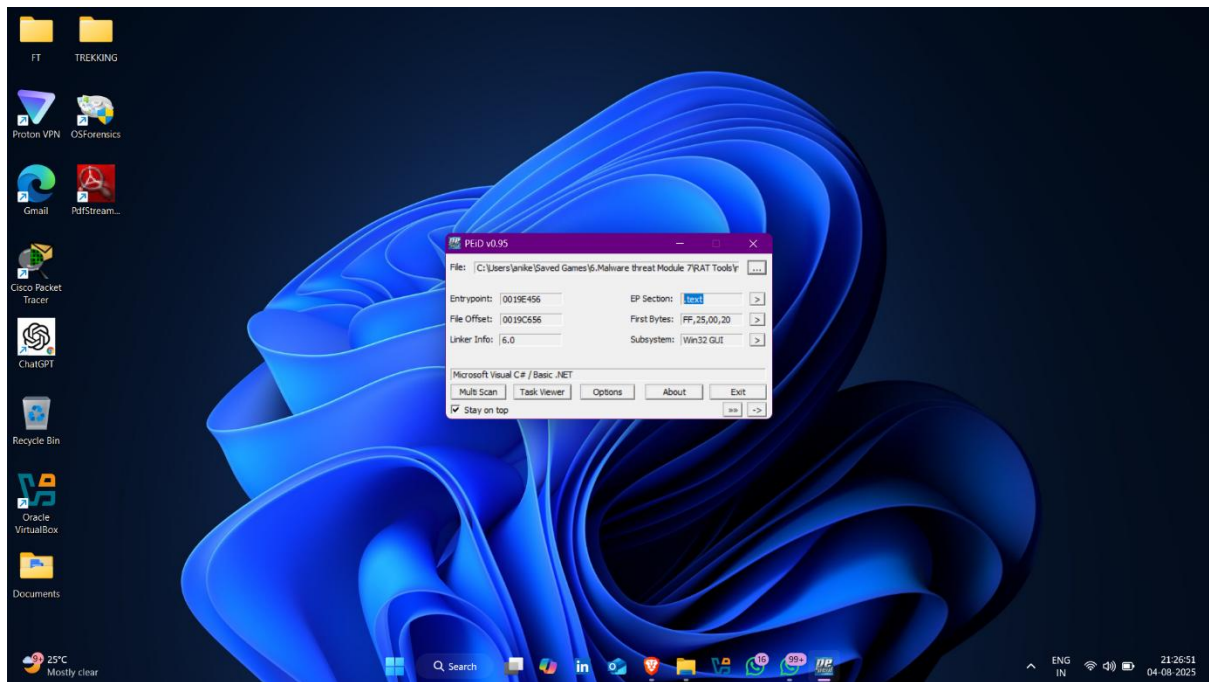


- Click on three dots (...)



- Select file and click on **open** option





B) Perform Static Analysis On A Suspicious File Using PEStudio

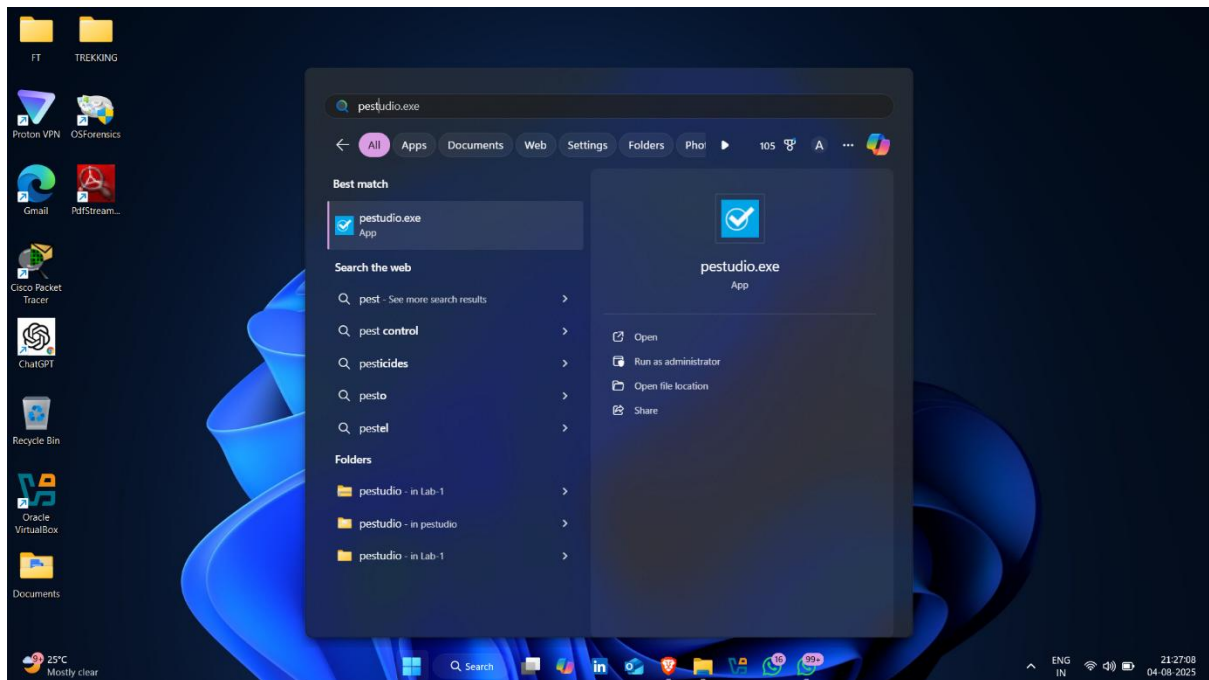
PEStudio is a **static analysis tool** for Windows executables that helps forensic investigators and malware analysts inspect a file **without executing it**, reducing the risk of infection.

Purpose

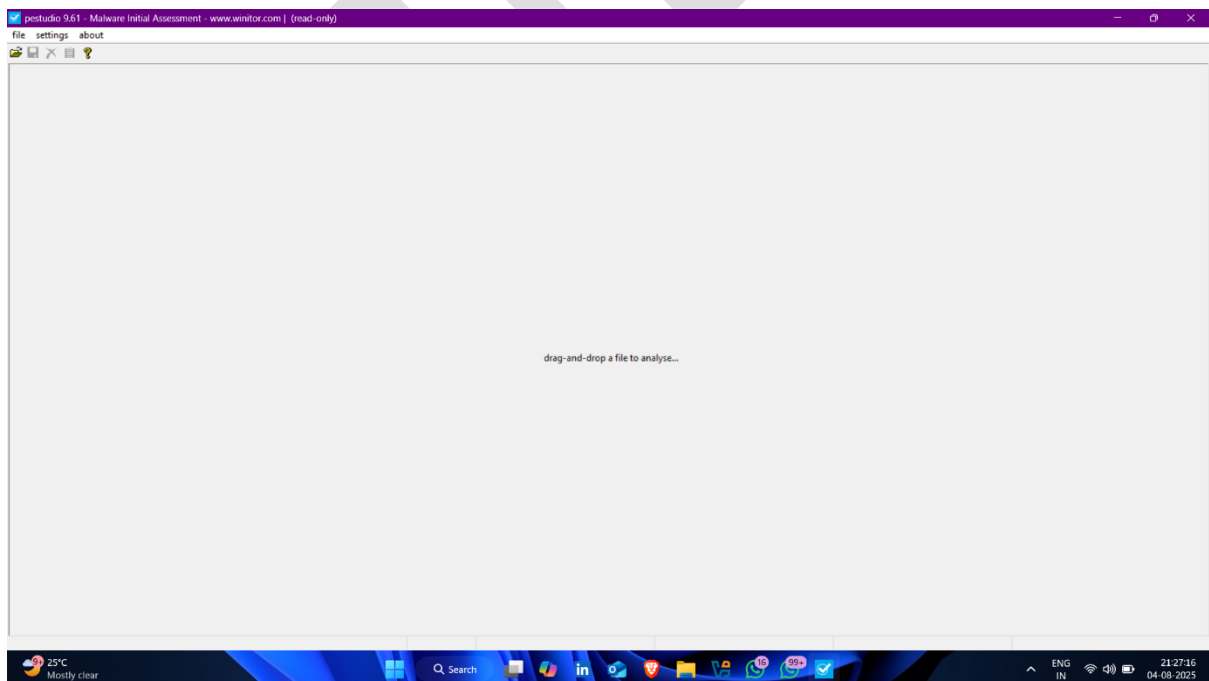
- To analyze **PE (Portable Executable)** files such as .exe, .dll, .sys.
- To identify **suspicious or malicious characteristics** before running the file.

How to use it -:

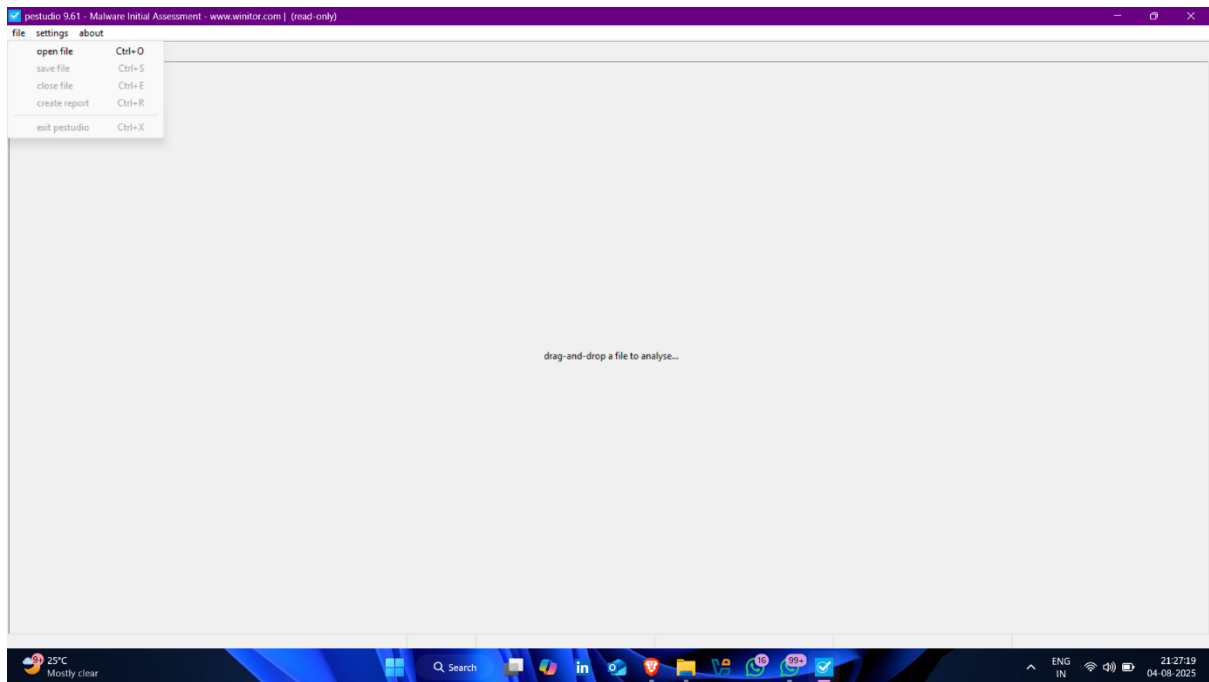
- Open PEStudio



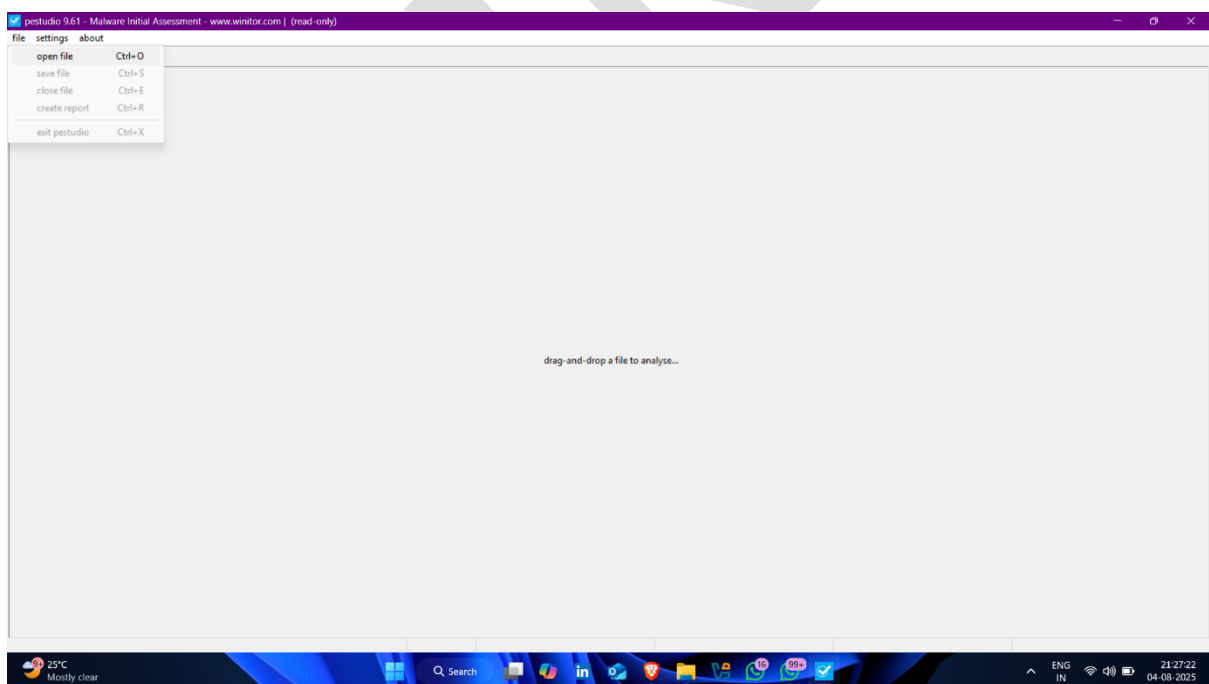
- PEStudio Started ☒



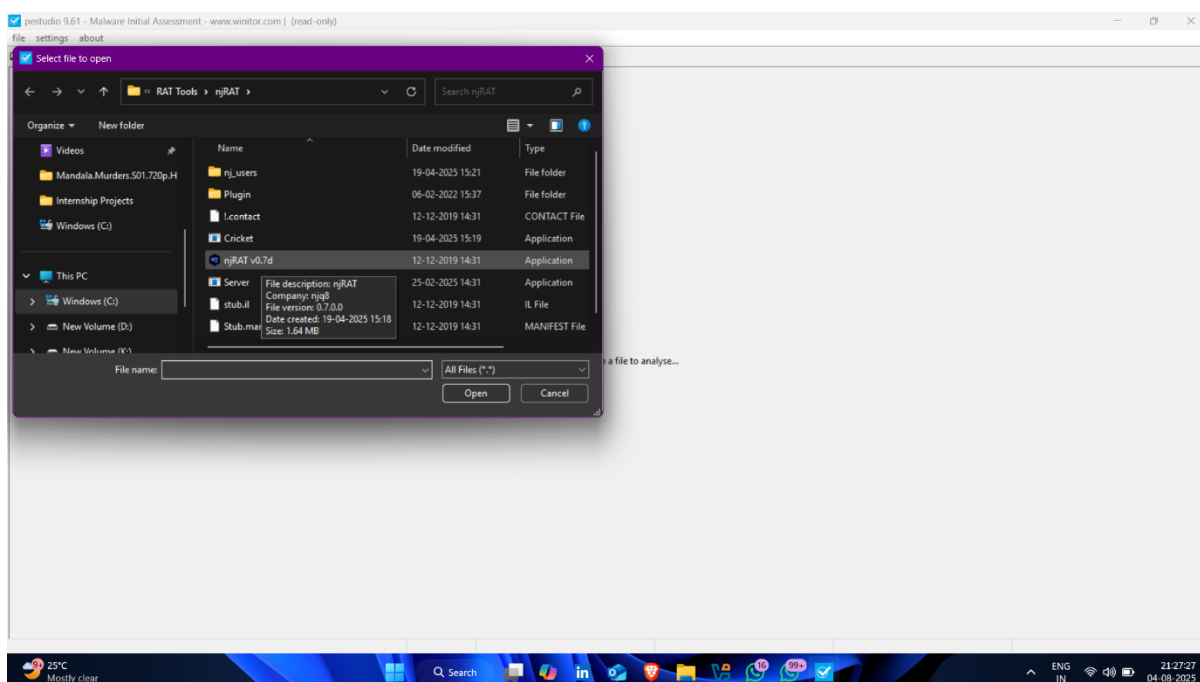
- Click on **file** option



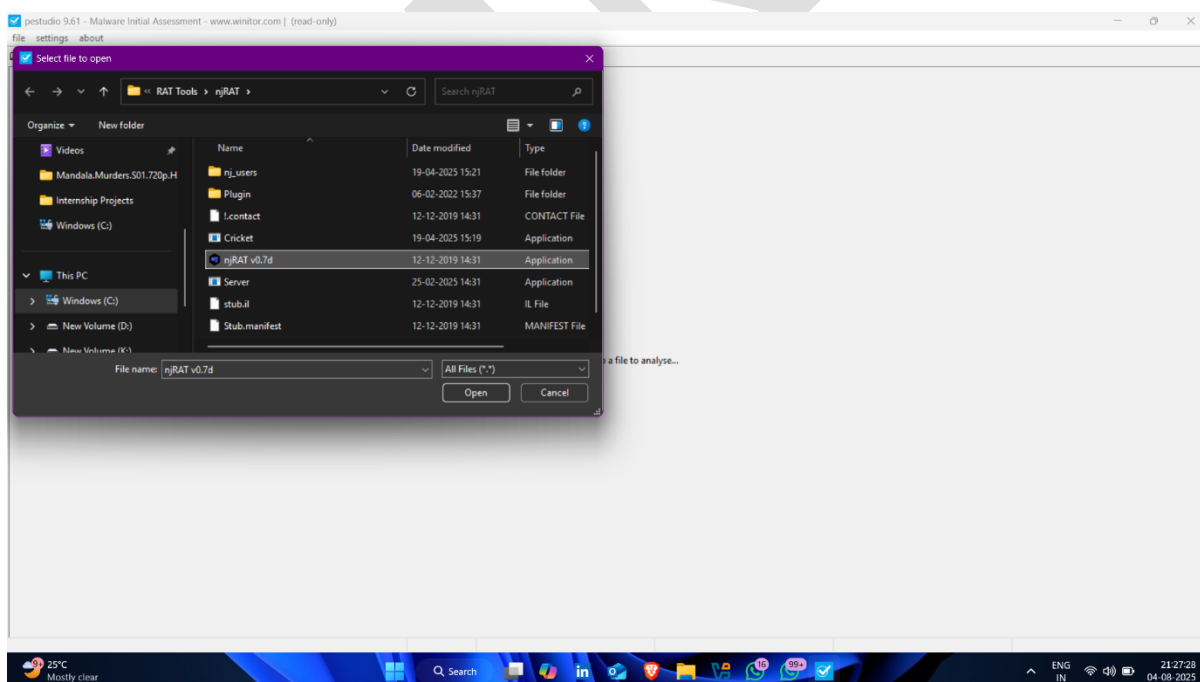
- Now click on **Open File** option



- **Select File**



- **Click on Open**



- Here, there are many inbuilt open source tools 🖱️

The screenshot shows the VirusTotal interface for a file named 'njrat.exe'. The file is located at 'c:\users\anki\save games\6.malware threat module 7\rat tools\njrat\njrat v0.7d.exe (read-only)'. The file properties are displayed in a table:

property	value
file	
file > sha256	BF853789B938BDC5DABAABE52511379A332C7CF238266A218FCB0318A62E85CB
file > first 32 bytes (hex)	4D 5A 90 00 03 00 00 04 00 00 FF 00 00 B8 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
file > first 32 bytes (text)	MZ.....@.....
file > info	size: 1723904 bytes, entropy: 5.250
file > type	executable, 32-bit, GUI
file > version	0.7.0.0
file > description	njRAT
#1 double-click > jump	FF 25 00 20 40 00
file > name	0x0019E456 (section, rsrc)
file > signature	Microsoft Linker 6.0
stamps	
stamp > compiler	Thu Dec 12 04:44:36 2013 (UTC)
stamp > debug	n/a
stamp > resource	n/a
stamp > import	n/a
stamp > export	n/a
names	
file > name	c:\users\anki\save games\6.malware threat module 7\rat tools\njrat\njrat v0.7d.exe
debug > file	n/a
export	n/a
version > original-file-name	njRAT.exe
manifest	MyApplication.app
.NET > module > name	njRAT.exe
certificate > program-name	n/a

- Virus Total Result 🖱️✅

The screenshot shows the VirusTotal interface for a file named 'njrat.exe'. The file is located at 'c:\users\anki\save games\6.malware threat module 7\rat tools\njrat\njrat v0.7d.exe (read-only)'. The detection results are displayed in a table:

vendor (72/72)	score (60/72)	date (dd.mm.yyyy)	age (days)
ALYac	Gen:Variant.Application.MSILPerseus.2112	25.07.2025	10
APEX	undetected	22.07.2025	13
AVG	MSL:Agent-DAG [Trj]	24.07.2025	11
Acronis	undetected	28.03.2024	494
AlmLab-V3	Trojan.Win32.Bladabindi.R101782	25.07.2025	10
Alibaba	VerTool.MSIL/Bladabindi.c79c368	27.05.2019	2361
Antiy-AVL	undetected	25.07.2025	10
Arcabit	Trojan.Application.MSILPerseus.D840	25.07.2025	10
Avast	MSL:Agent-DAG [Trj]	24.07.2025	11
Avira	HEUR/AGEN.1324488	25.07.2025	10
Baidu	MSIL.Trojan.Bladabindi.n	18.03.2019	2331
BitDefender	Gen:Variant.Application.MSILPerseus.2112	25.07.2025	10
Bkav	undetected	25.07.2025	10
CAT-QuickHeal	Trojan.Generic.FCS12095946	24.07.2025	11
CMC	undetected	25.07.2025	10
CTX	exe.trojan.msil	25.07.2025	10
ClamAV	Win.Malware.Agent-6352945-0	24.07.2025	11
CrowdStrike	win/malicious_confidence_100% (W)	26.10.2023	648
Cyforce	Unsafe	10.07.2025	25
Cynet	undetected	24.07.2025	11
DeepInstinct	MALICIOUS	24.07.2025	11
DryWeb	BackDoor.Bladabindi.13385	25.07.2025	10
ESET-NOD32	MSIL/Bladabindi.AN	25.07.2025	10
Elastic	malicious (high confidence)	24.07.2025	11
Emisoft	Gen:Variant.Application.MSILPerseus.2112 (B)	25.07.2025	10
F-Secure	Heuristic.HEUR/AGEN.1324488	25.07.2025	10
Fortinet	MSIL/Generic.DN.22E306tr	25.07.2025	10
GData	Gen:Variant.Application.MSILPerseus.2112	25.07.2025	10
Google	Detected	25.07.2025	10
Gridinsoft	Trojan.Win32.Agent.dg	25.07.2025	10
Ikarus	Trojan.Bladabindi	24.07.2025	11
Jiangmin	Trojan.MSIL.ammw	24.07.2025	11
K7AntiVirus	Trojan (700000201)	25.07.2025	10
K7GW	Trojan (700000201)	25.07.2025	10
Kaspersky	HEUR:Trojan.Win32.Generic	25.07.2025	10
Kingsoft	Win32.Trojan.Generic.a	24.07.2025	11
Lionic	Trojan.Win32.Bladabindi.4lc	24.07.2025	11

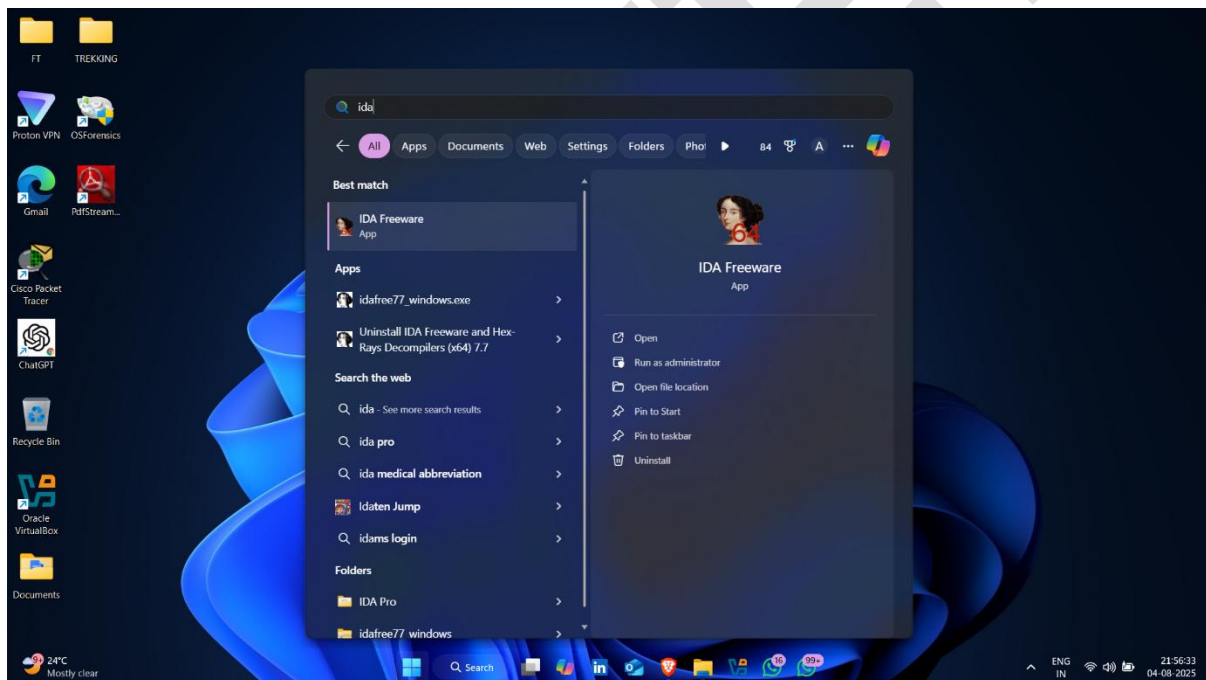
C) Perform Static Analysis On A Suspicious File Using IDA Pro

IDA Pro (Interactive DisAssembler Professional) is one of the most popular **reverse engineering tools** used in malware analysis, vulnerability research, and software forensics.

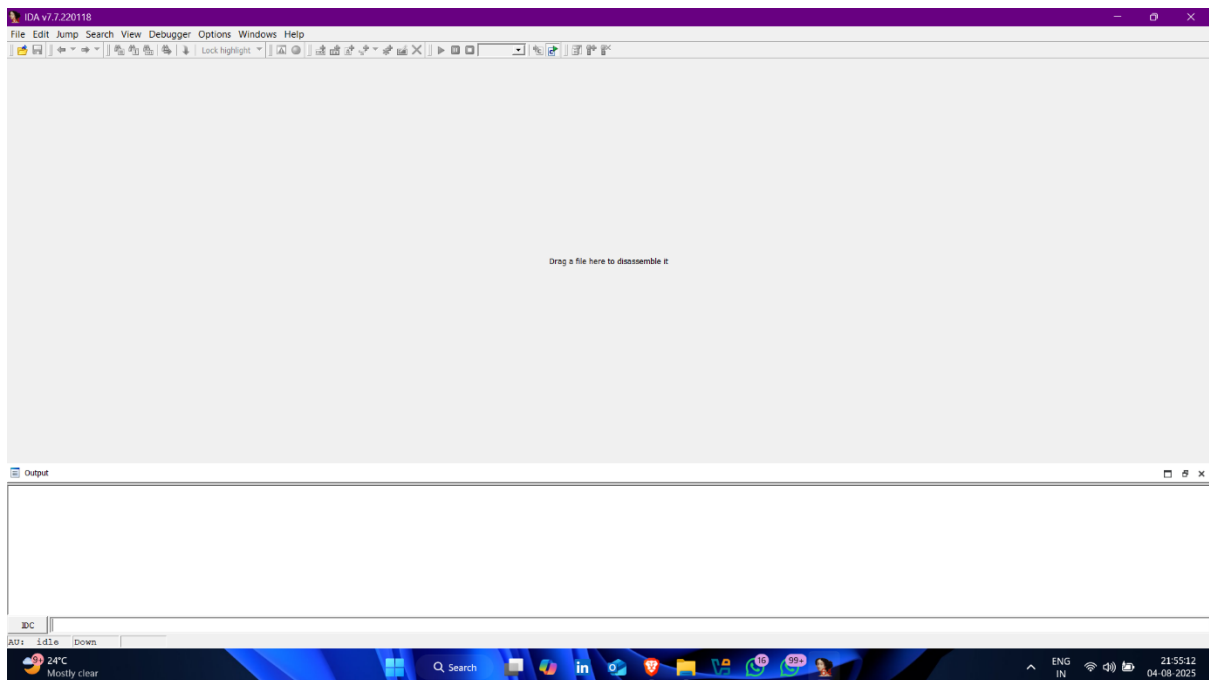
It is mainly used to **disassemble** compiled programs (turn machine code into assembly code) and help analysts understand how the program works — **without having the source code**.

How to use it -:

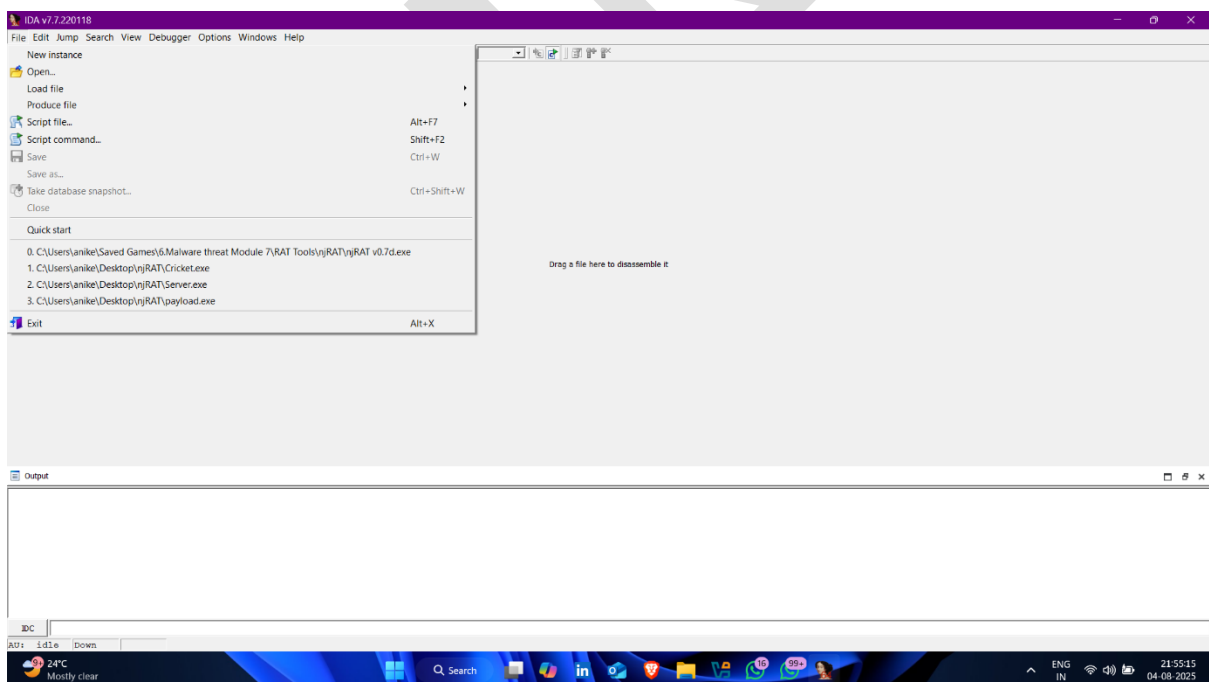
- Open IDA Freeware application



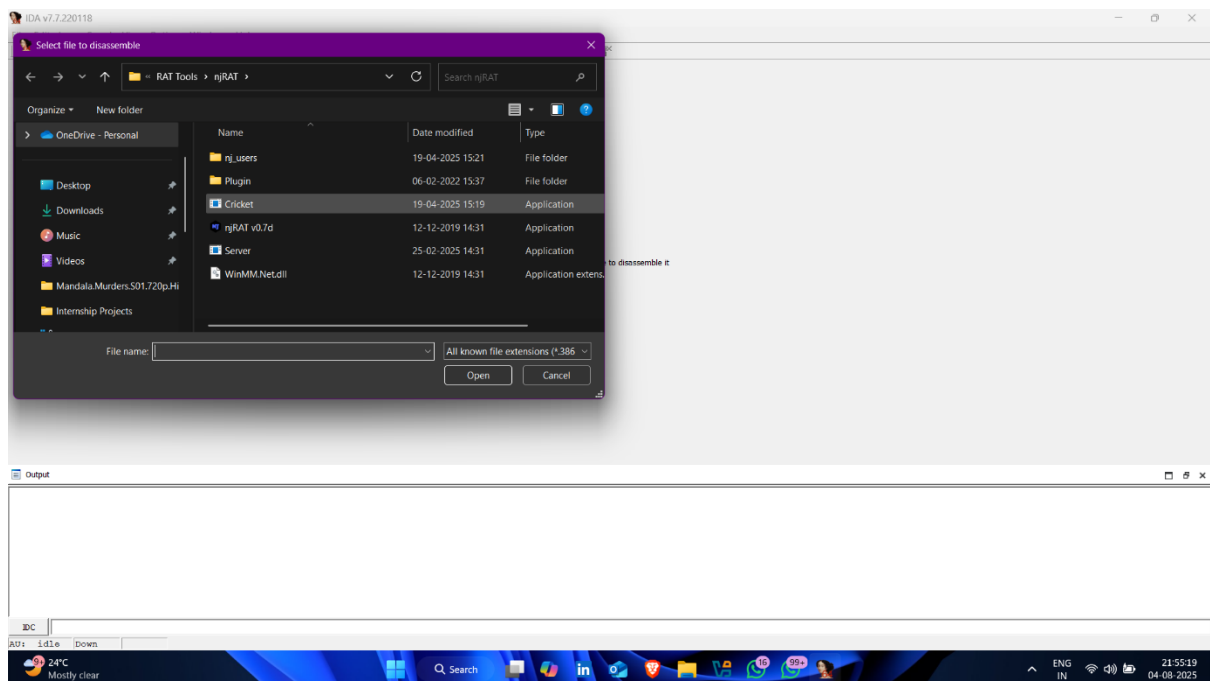
- Application Interface 🖱️



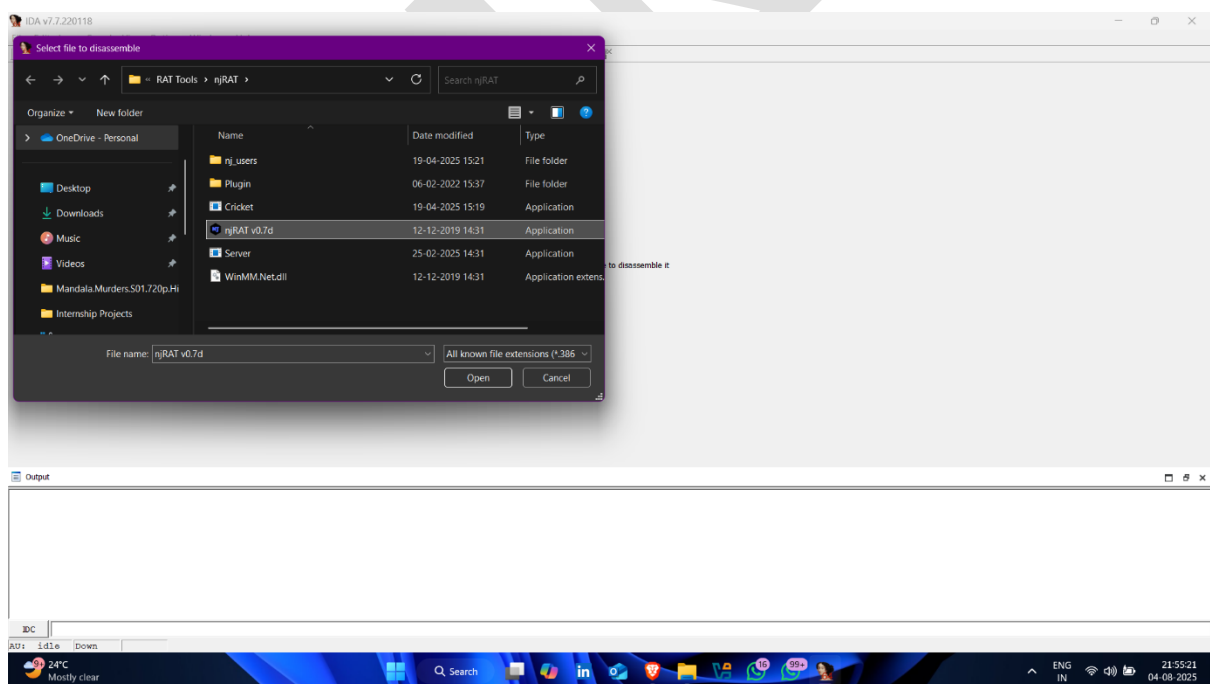
- Click on File Section and then click on open option



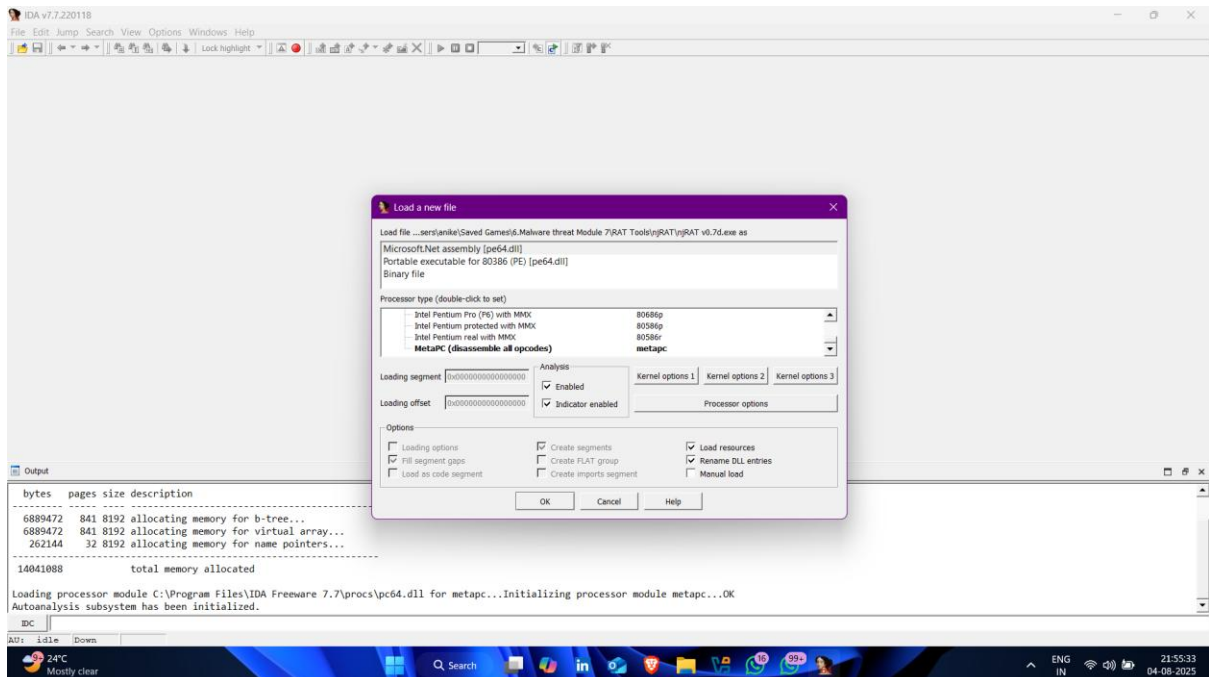
- **Select File**



- **After select file click on open**



- Now check two options – 1) load Resources and 2) Rename DLL Entries and then click on ok



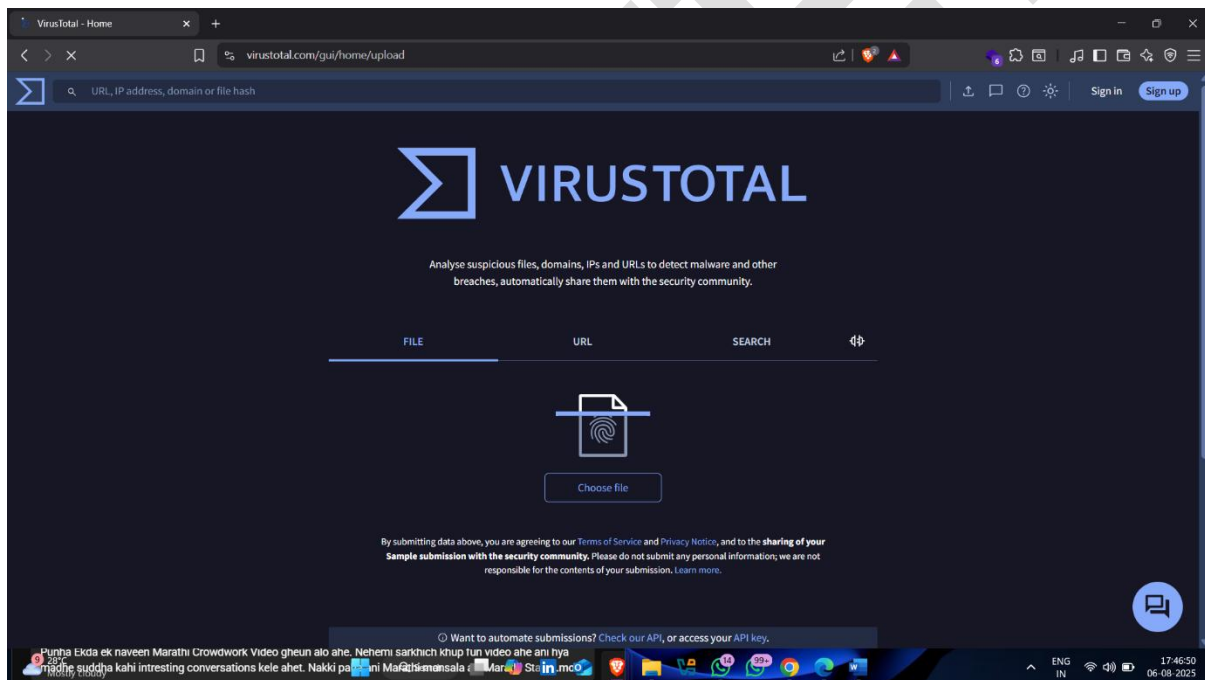
Lab-2 Perform Static Analysis On A Suspicious File Using Virus Total

VirusTotal is a **free online malware scanning and threat intelligence platform** that analyzes suspicious files, URLs, IP addresses, and domains to detect potential malicious activity.

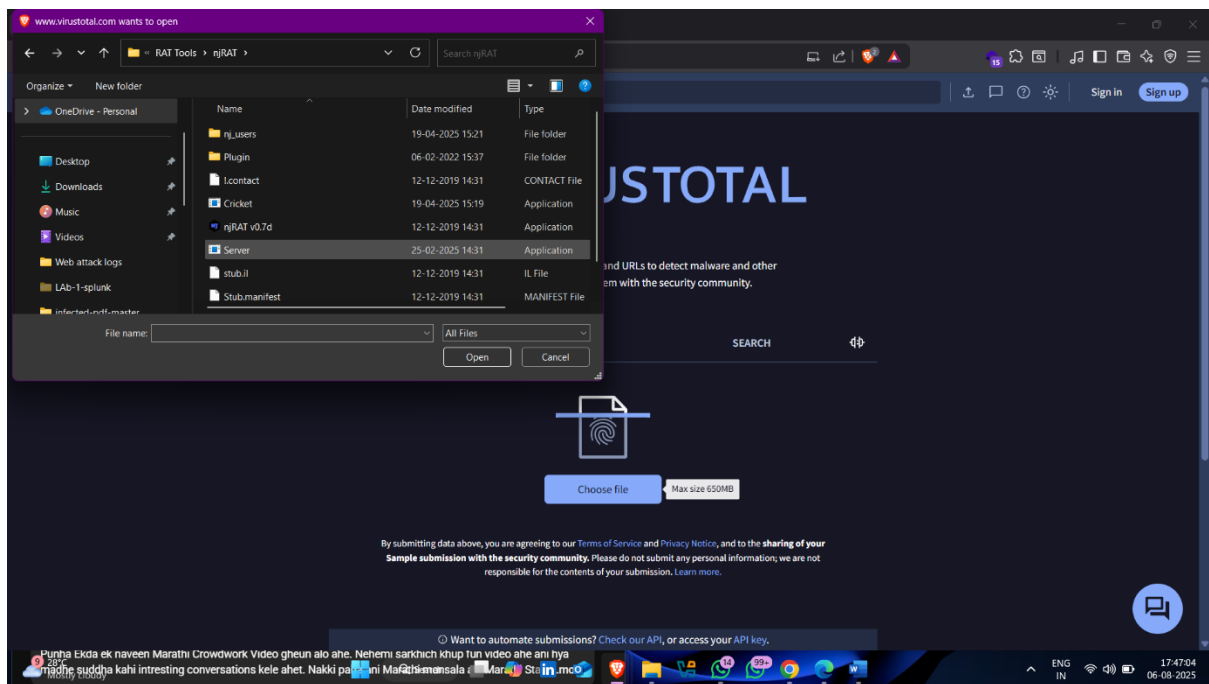
It works by scanning the submitted file or link with **multiple antivirus engines** and security tools at the same time, then showing you the results in one report.

How to use it :-

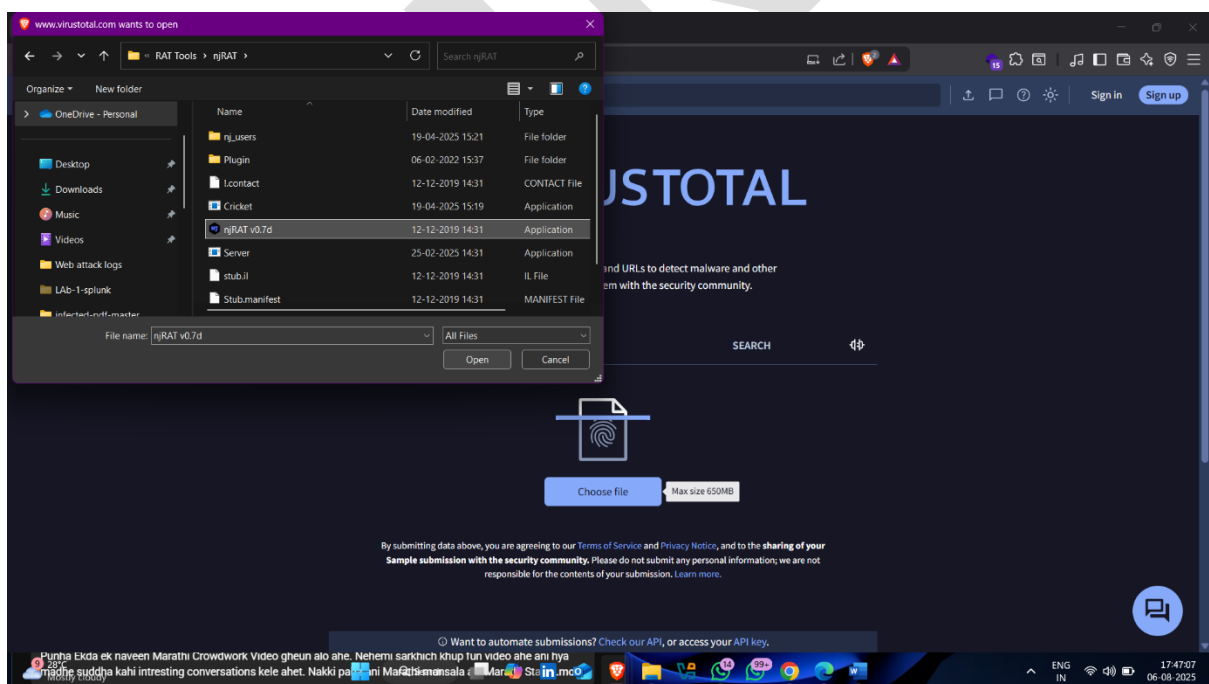
- Open Virus Total Website and click on **Choose File**



- **Select File**



- **Click on open**



- Here result – malicious File 🖱️ ✅

Max size 650MB

virustotal.com/gui/file/bf853789b938bd5da8aeb52511379a332c7cf238266a21bfc0318a62e85cb

bf853789b938bd5da8aeb52511379a332c7cf238266a21bfc0318a62e85cb

62 / 72
Community Score 47

62/72 security vendors flagged this file as malicious

Reanalyze Similar More

bf853789b938bd5da8aeb52511379a332c7cf238266a21bfc0318a62e85cb
njRAT.exe

Size 1.64 MB Last Analysis Date 1 day ago

peexe idle via-tor runtime-modules direct-cpu-clock-access assembly detect-debug-environment

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY 22+

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label trojan.msil.bladabindi Threat categories trojan Family labels msil bladabindi msilperseus

Security vendors' analysis

Vendor	Detection
AhnLab-V3	Trojan.Win32.Bladabindi.R101782
AllCloud	Trojan:Win/Perseus.AZ
Arcabit	Trojan.Application.MSILPerseus.D840
Avast	MSIL:Agent-DAG [Trj]
Avira (no cloud)	HEUR/AGEN.1324488
Alibaba	VirTool:MSIL/Bladabindi.c73c5368
ALYac	Gen:Variant.Application.MSILPerseus.2112
Arctic Wolf	Unsafe
AVG	MSIL:Agent-DAG [Trj]
Baidu	MSIL.Trojan.Bladabindi.n

Do you want to automate checks?

Punha Ekda ek naveen Marathi Crowdwork Video gheun alo ahe. Nehem sarkhich khup tun video ahe ani hya
 Here suddha kahi intresting conversations kele ahet. Nakki pa... ni Maf...nansala... Mar... Stu... mco

ENG IN 17:48:25 06-08-2025

Lab-3 Forensics Examination Of Suspicious PDF File

Forensics Examination of a Suspicious PDF File means analyzing a PDF document to check whether it is **malicious, tampered, or contains hidden/embedded harmful content** such as malware, phishing links, or exploits.

Attackers often use **PDF files** for malware delivery because PDFs can contain:

- Embedded JavaScript code
- Malicious macros
- Exploit payloads
- Links to phishing sites

Goals of the Examination

1. **Verify authenticity** – check if the file is genuine or altered.
2. **Detect hidden threats** – malware, embedded scripts, hidden files.
3. **Identify attack vectors** – exploit vulnerabilities in Adobe Reader or other PDF viewers.
4. **Gather evidence** – for cybercrime investigation.

Steps in PDF Forensics Examination

1. Initial Analysis (Static Analysis)

- **Check file metadata:** creator, creation date, modification date.
 - Tools: pdftinfo (Linux), ExifTool, PDF-Analyzer.
- **Hashing:** generate MD5/SHA256 for evidence.
- **Check file structure:**
 - PDFs are made of objects (text, images, scripts). Look for **extra or suspicious objects**.
- **Look for suspicious elements:**

- /JS or /JavaScript – indicates embedded scripts.
 - /OpenAction – triggers something when the file opens.
 - /AA (Additional Actions) – can run scripts automatically.
-

2. Suspicious Content Extraction

- **Extract JavaScript or embedded files:**

- Tool: pdf-parser.py (by Didier Stevens).

Example:

pdf-parser.py suspicious.pdf

pdf-parser.py -s /JavaScript suspicious.pdf

- **Extract embedded files** (like EXE or ZIP):

- Tool: peepdf, pdfid, pdfdetach.
-

3. Behavioral Analysis (Dynamic Analysis)

- Open the PDF in a **sandbox** (isolated environment).
 - Monitor:
 - Network connections.
 - Processes started.
 - Files created/modified.
 - Tools: Cuckoo Sandbox, Joe Sandbox.
-

4. Check with Threat Intelligence

- Upload to **VirusTotal** or **Hybrid Analysis**.
 - See if the PDF matches known malware signatures.
-

Common Indicators of Malicious PDF

- Contains /JavaScript and obfuscated code.

- Has /Launch action to run commands.
- Embedded .exe or .vbs files.
- Links to phishing sites.
- Suspicious or mismatched creation dates.

A) Forensics Examination Of Suspicious PDF File Using DidierStevenSuite

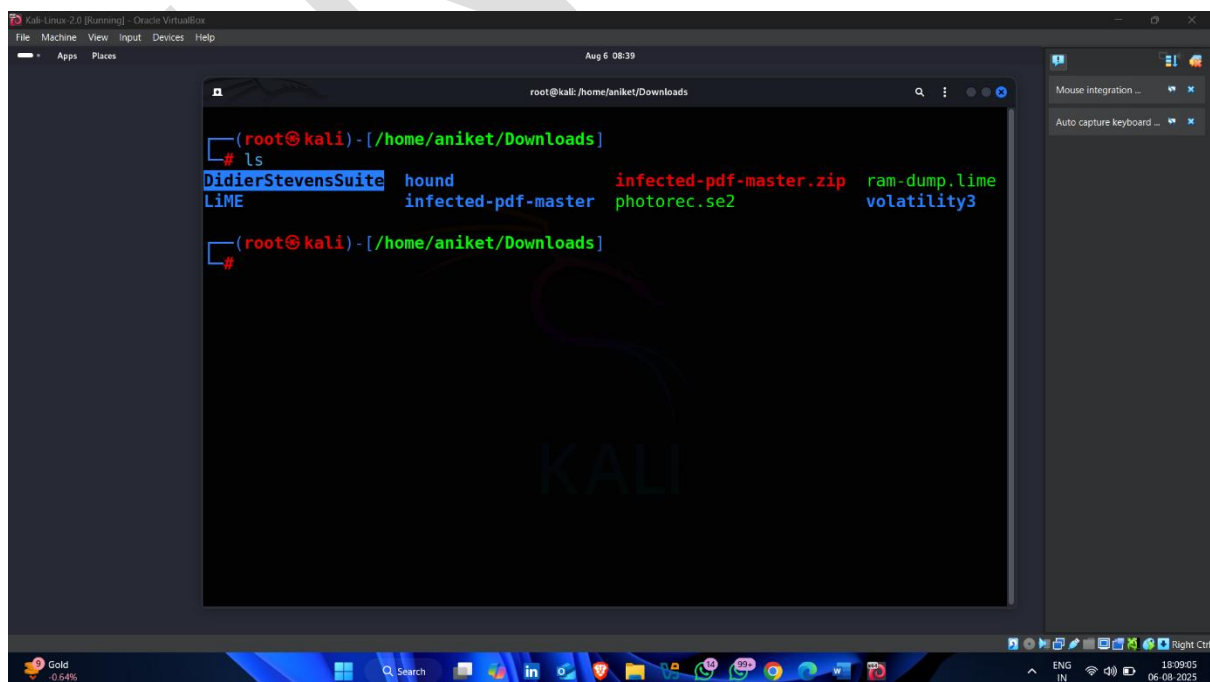
Didier Stevens Suite is a collection of **free, open-source forensic and malware analysis tools** created by **Didier Stevens**, a well-known security researcher.

These tools are especially useful for **PDF forensics, malicious document analysis, and general malware investigation.**

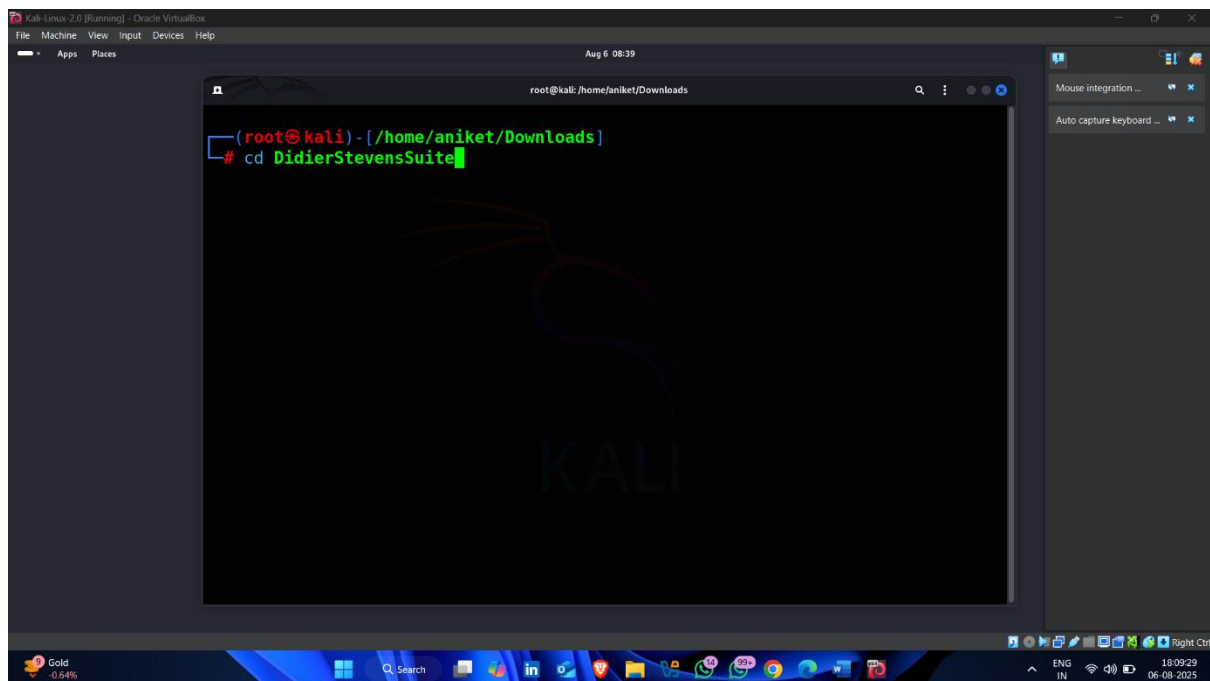
They are mostly written in **Python** and are designed to help analysts extract, parse, and examine potentially malicious content.

How to use it -:

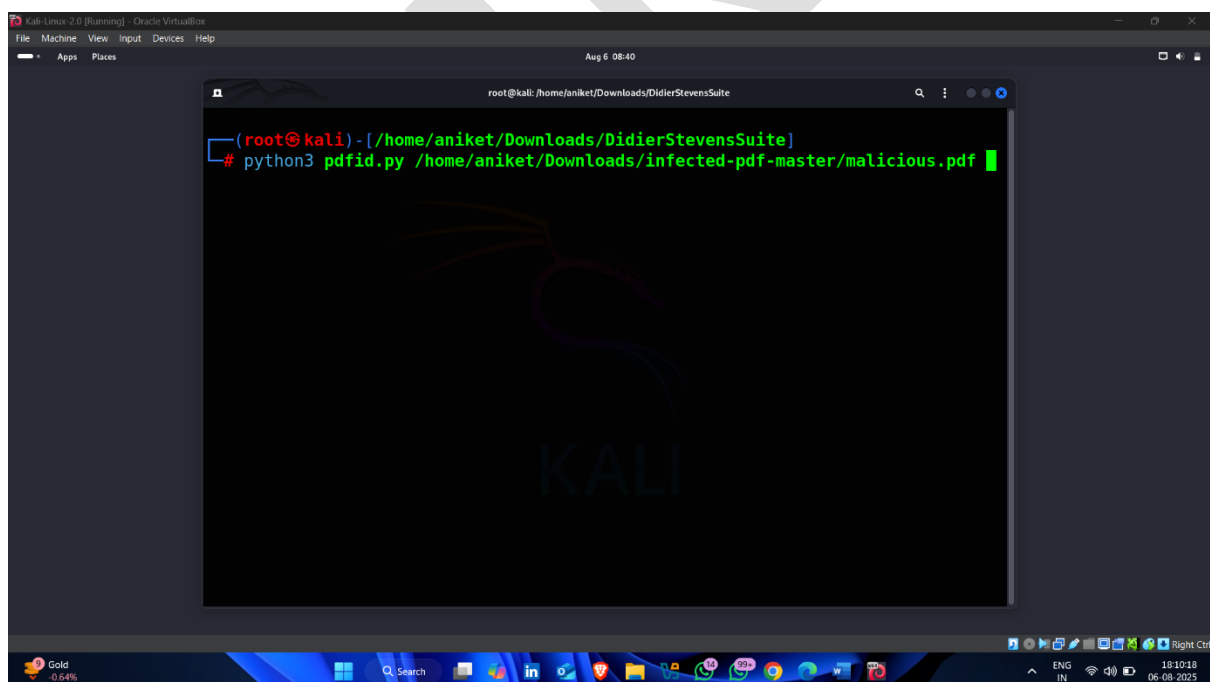
- Open **DidierStevenSuite** Directory



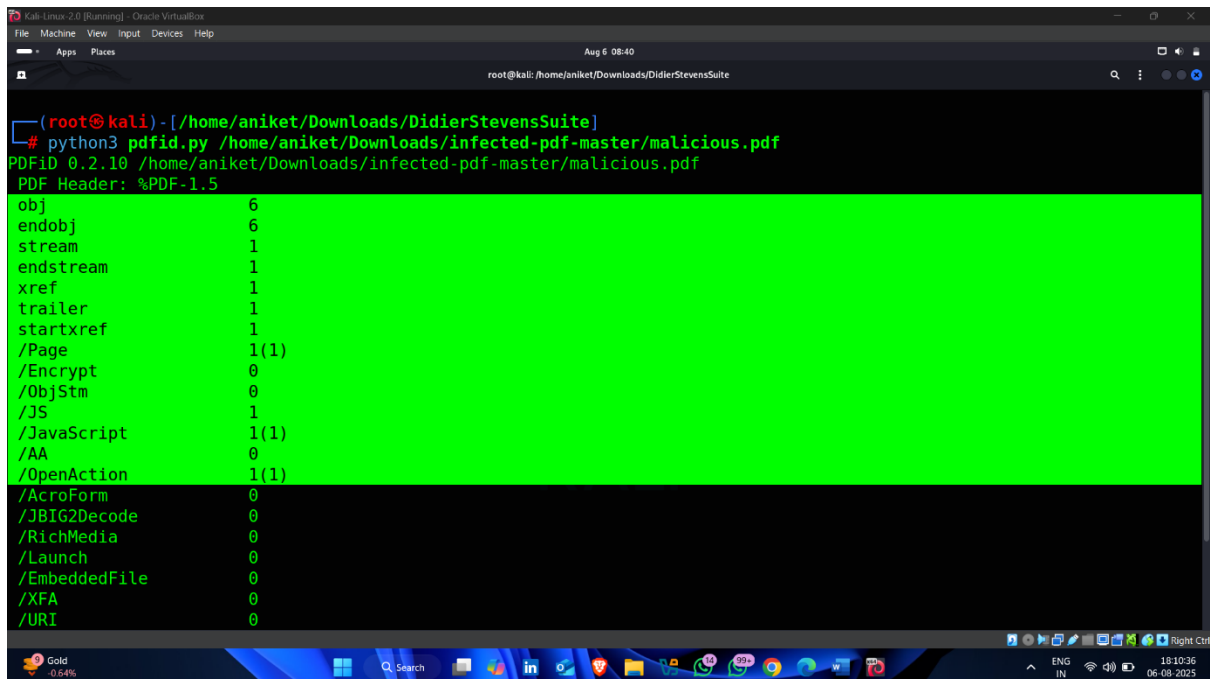
- Using **Cd DidierStevenSuite**



- Use Command - : `python3 pdfid.py < malicious pdf file path>`

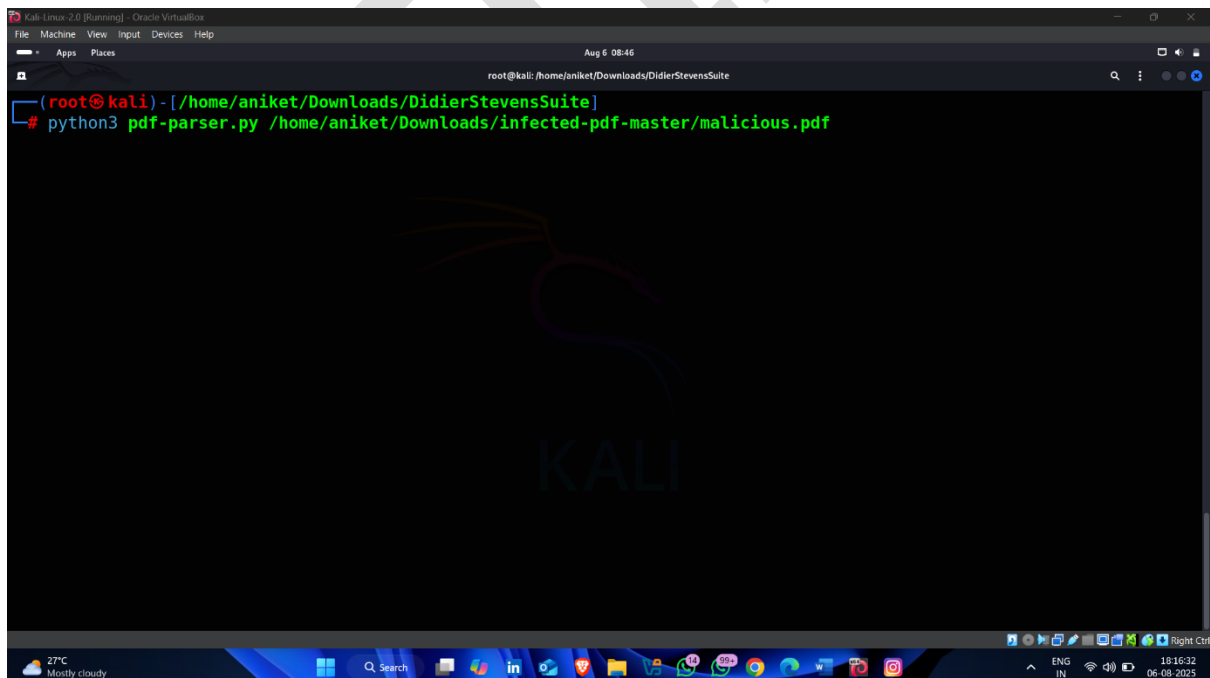


- Result --



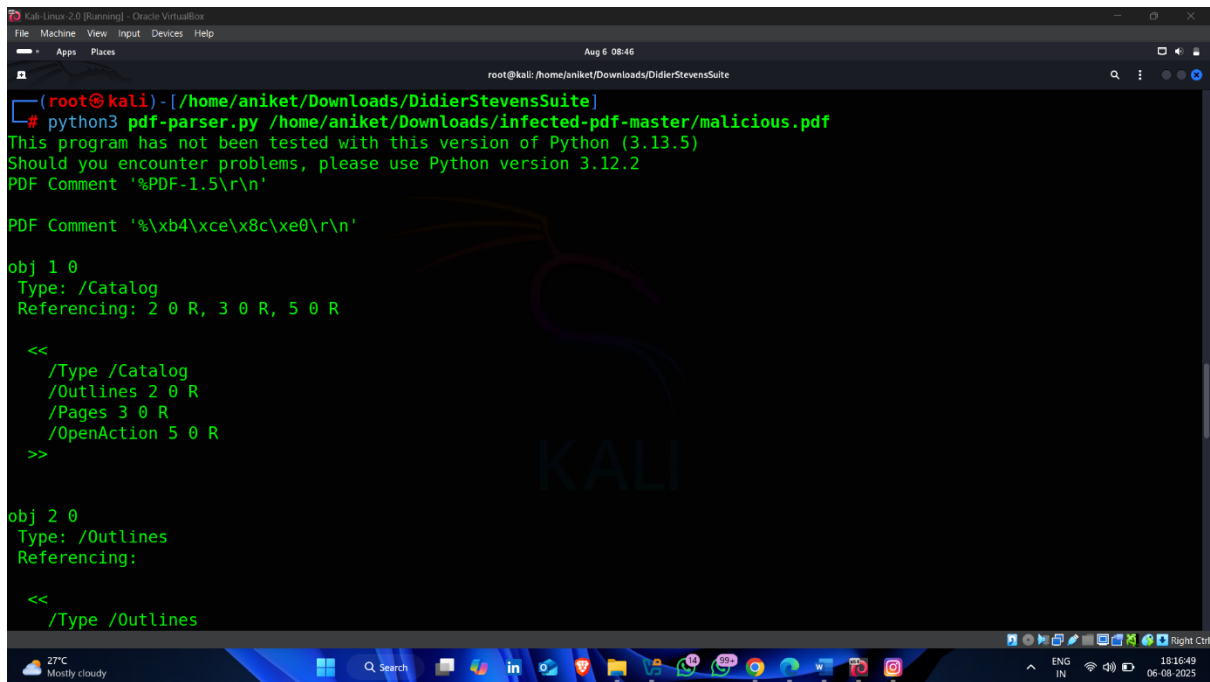
```
(root@kali) - [/home/aniket/Downloads/DidierStevensSuite]
# python3 pdfid.py /home/aniket/Downloads/infected-pdf-master/malicious.pdf
PDFID 0.2.10 /home/aniket/Downloads/infected-pdf-master/malicious.pdf
PDF Header: %PDF-1.5
obj 6
endobj 6
stream 1
endstream 1
xref 1
trailer 1
startxref 1
/Page 1(1)
/Encrypt 0
/ObjStm 0
/JS 1
/JavaScript 1(1)
/AA 0
/OpenAction 1(1)
/AcroForm 0
/JBIG2Decode 0
/RichMedia 0
/Launch 0
/EmbeddedFile 0
/XFA 0
/URI 0
```

- Use next command -: python3 pdf-parser.py <malicious pdf file path>



```
(root@kali) - [/home/aniket/Downloads/DidierStevensSuite]
# python3 pdf-parser.py /home/aniket/Downloads/infected-pdf-master/malicious.pdf
```

- Result --



```
(root@kali) - [~/home/aniket/Downloads/DidierStevensSuite]
# python3 pdf-parser.py /home/aniket/Downloads/infected-pdf-master/malicious.pdf
This program has not been tested with this version of Python (3.13.5)
Should you encounter problems, please use Python version 3.12.2
PDF Comment '%PDF-1.5\r\n'

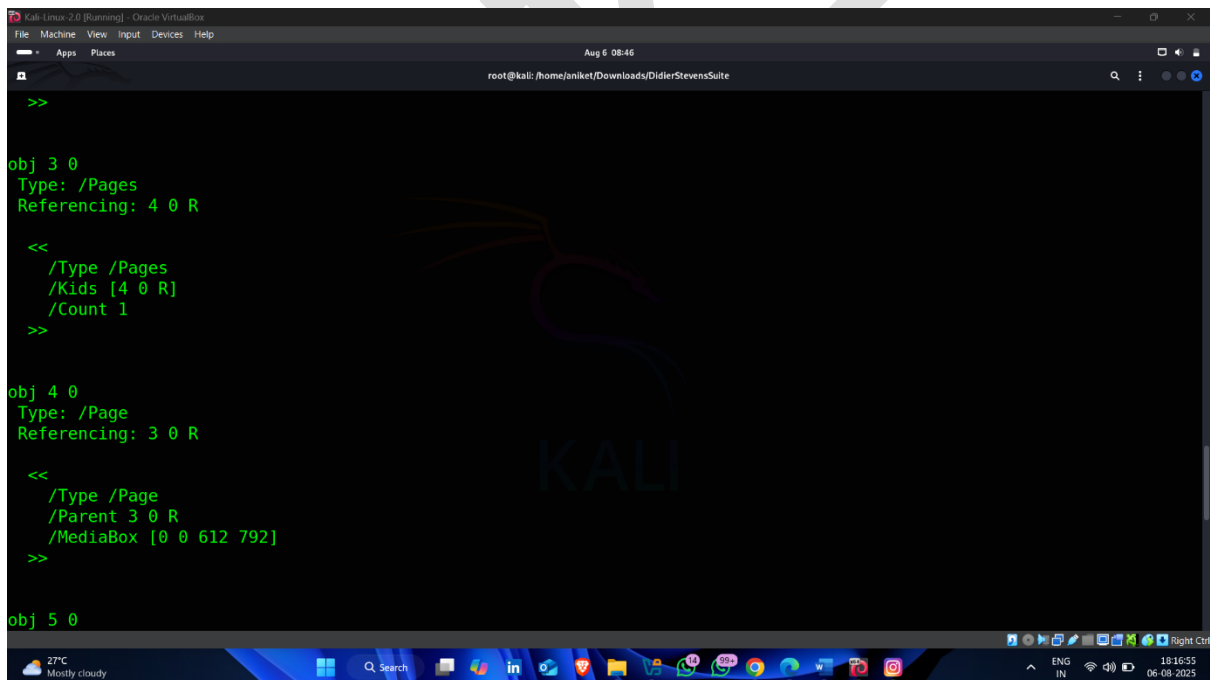
PDF Comment '%\xb4\xce\x8c\xe0\r\n'

obj 1 0
Type: /Catalog
Referencing: 2 0 R, 3 0 R, 5 0 R

<<
  /Type /Catalog
  /Outlines 2 0 R
  /Pages 3 0 R
  /OpenAction 5 0 R
>>

obj 2 0
Type: /Outlines
Referencing:

<<
  /Type /Outlines
```



```
>>

obj 3 0
Type: /Pages
Referencing: 4 0 R

<<
  /Type /Pages
  /Kids [4 0 R]
  /Count 1
>>

obj 4 0
Type: /Page
Referencing: 3 0 R

<<
  /Type /Page
  /Parent 3 0 R
  /MediaBox [0 0 612 792]
>>

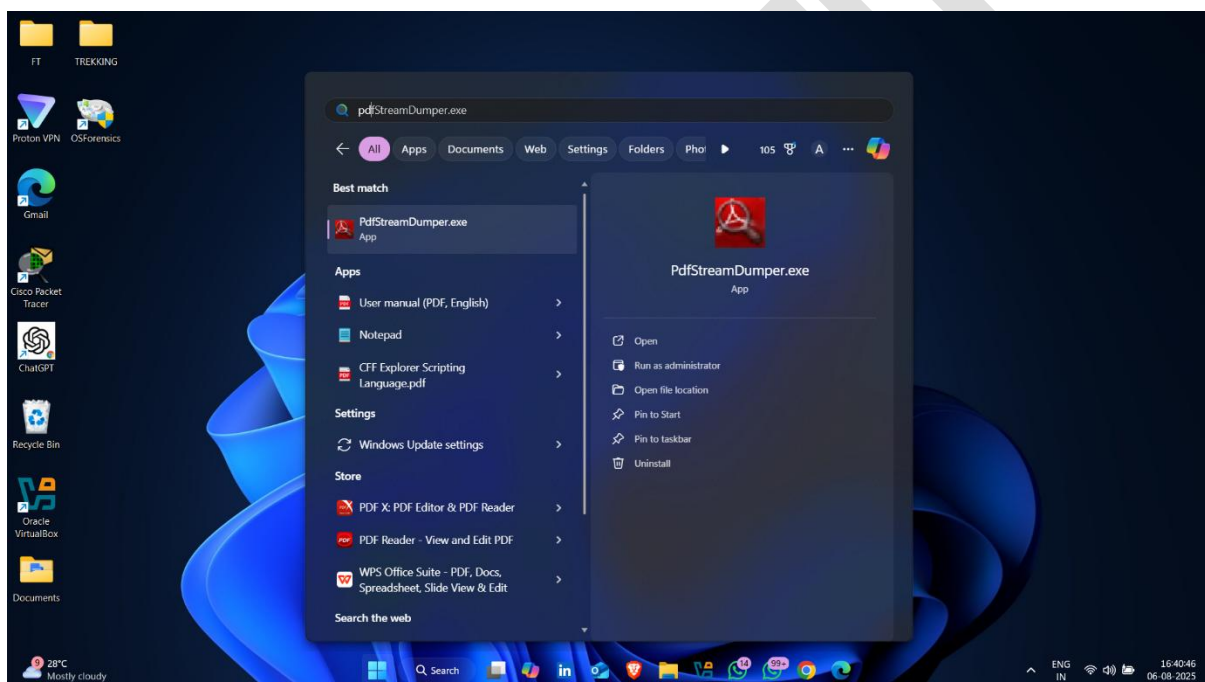
obj 5 0
```

B) Forensics Examination Of Suspicious PDF File Using Pdf-Streamer-Dumper

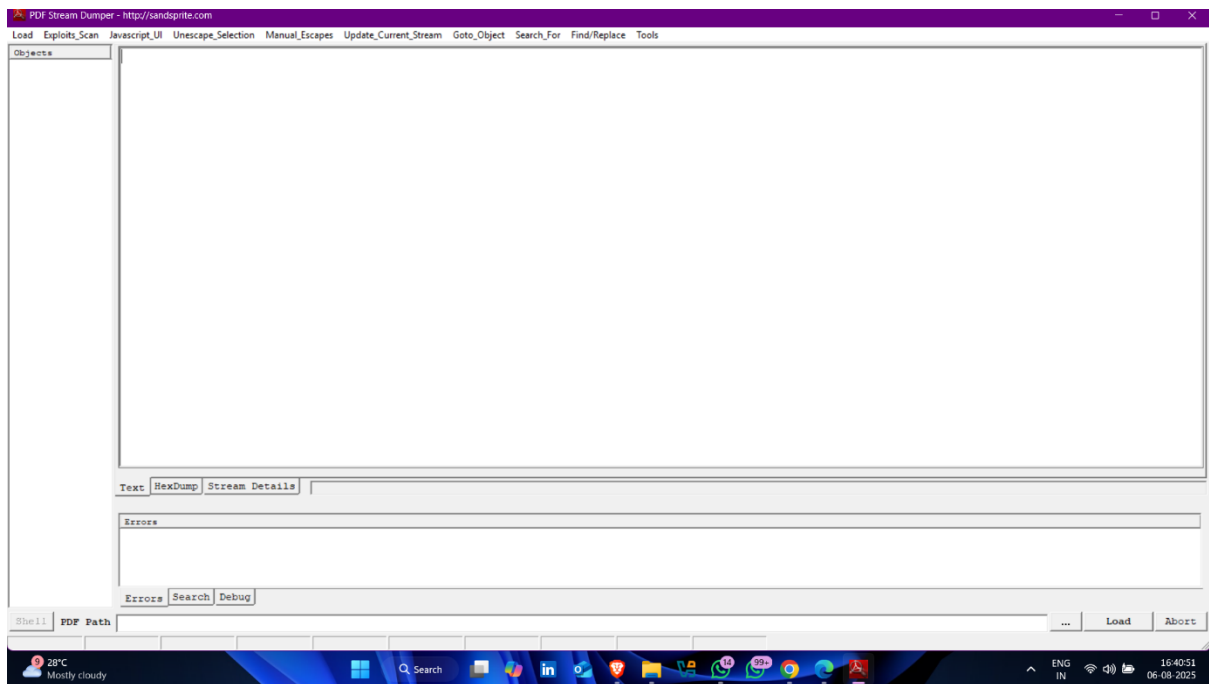
PDF Stream Dumper (also known as **PDFStreamDumper**) is a powerful, free GUI-based tool used in **PDF forensics** to analyze and extract content from suspicious PDF files — especially ones with embedded **JavaScript** or exploit code. It helps analysts uncover malicious payloads and vulnerabilities in PDF streams.

How to use it -:

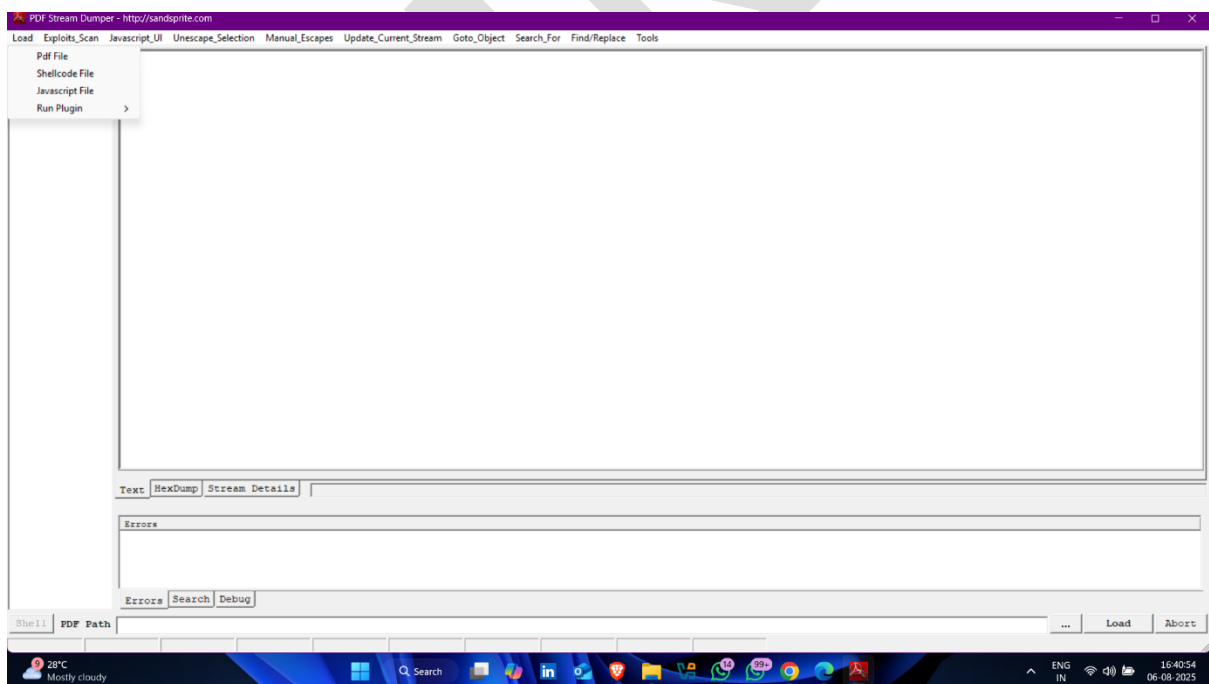
- Open **pdf-Stream-Dumper**



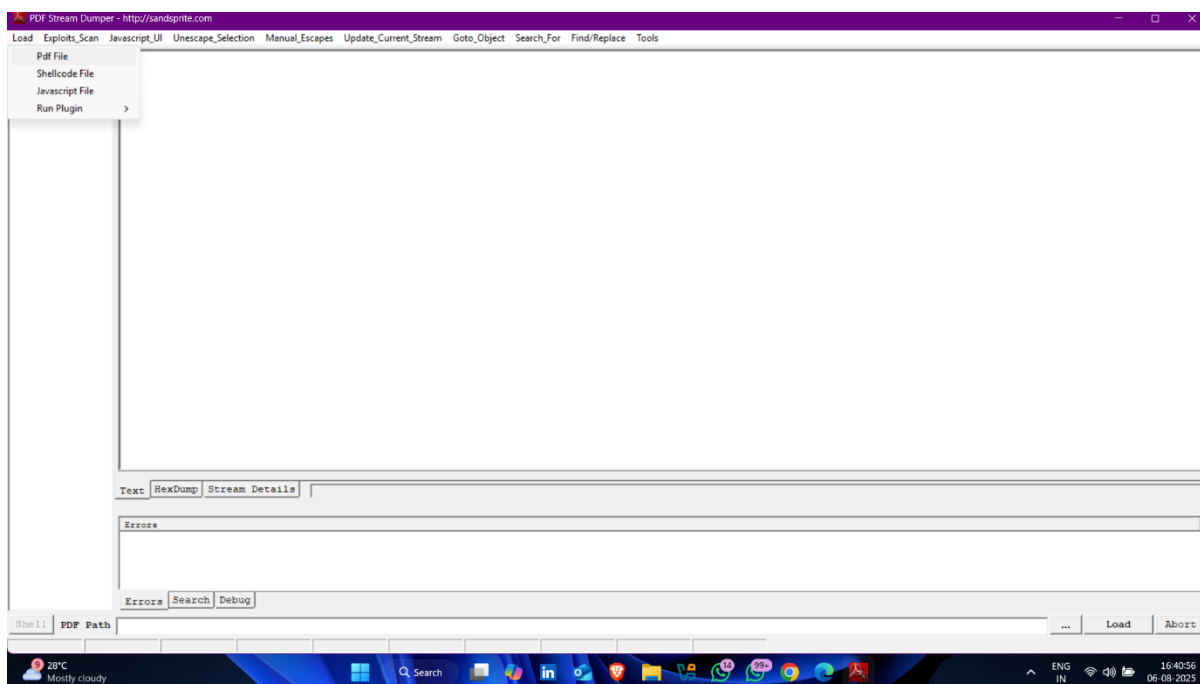
- **Application open**



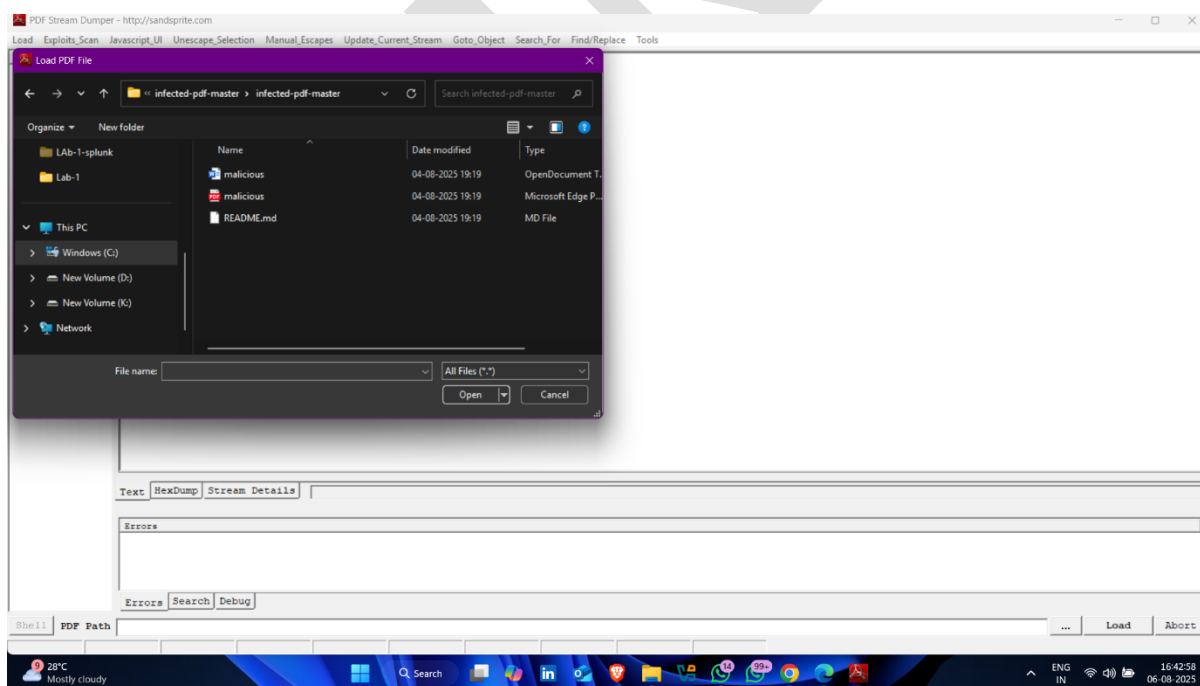
- **Now click on Load option**



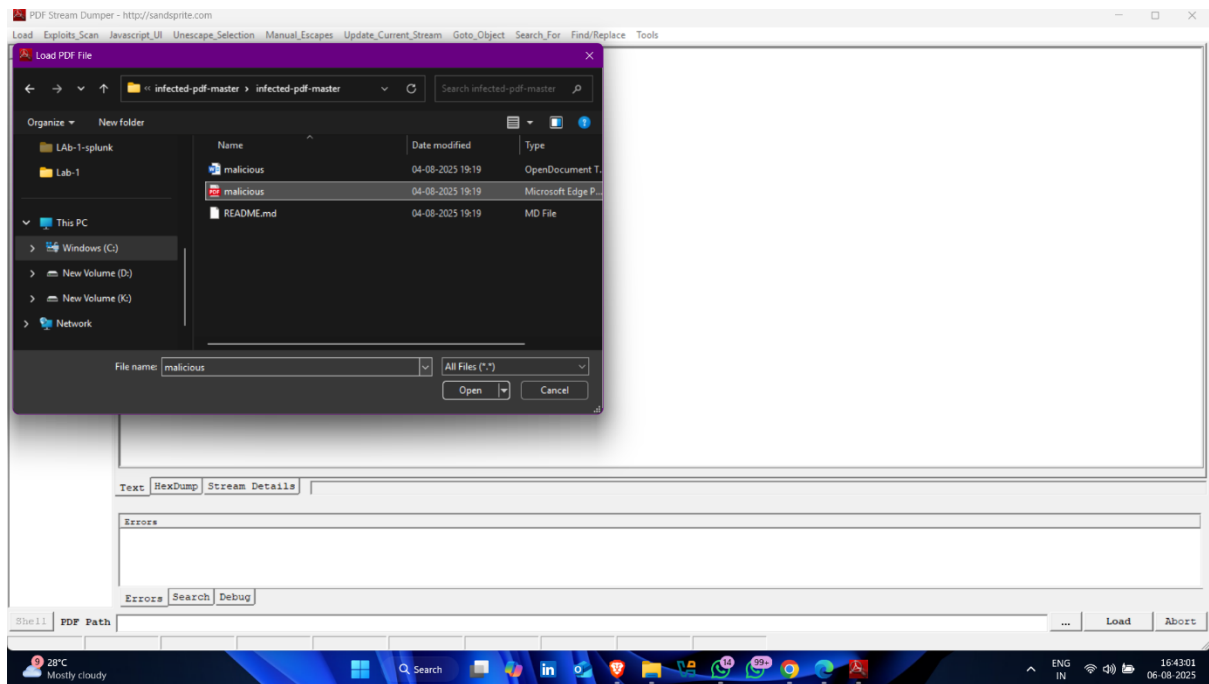
- Then click on **pdf file**



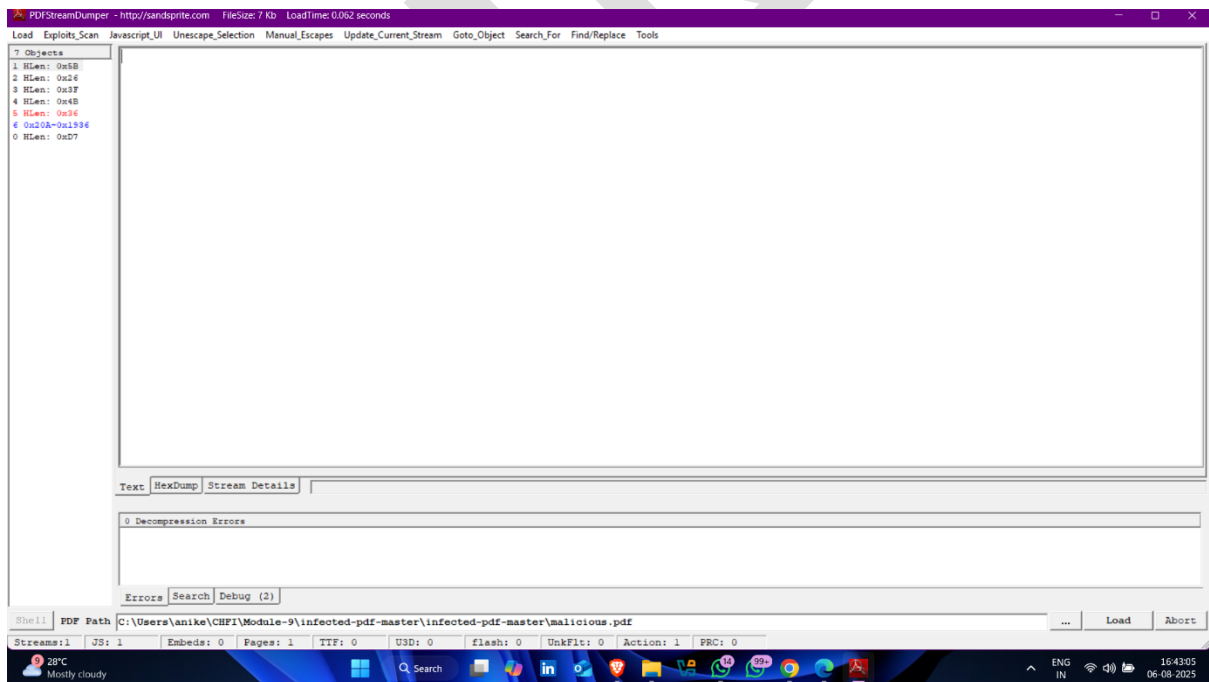
- Select file



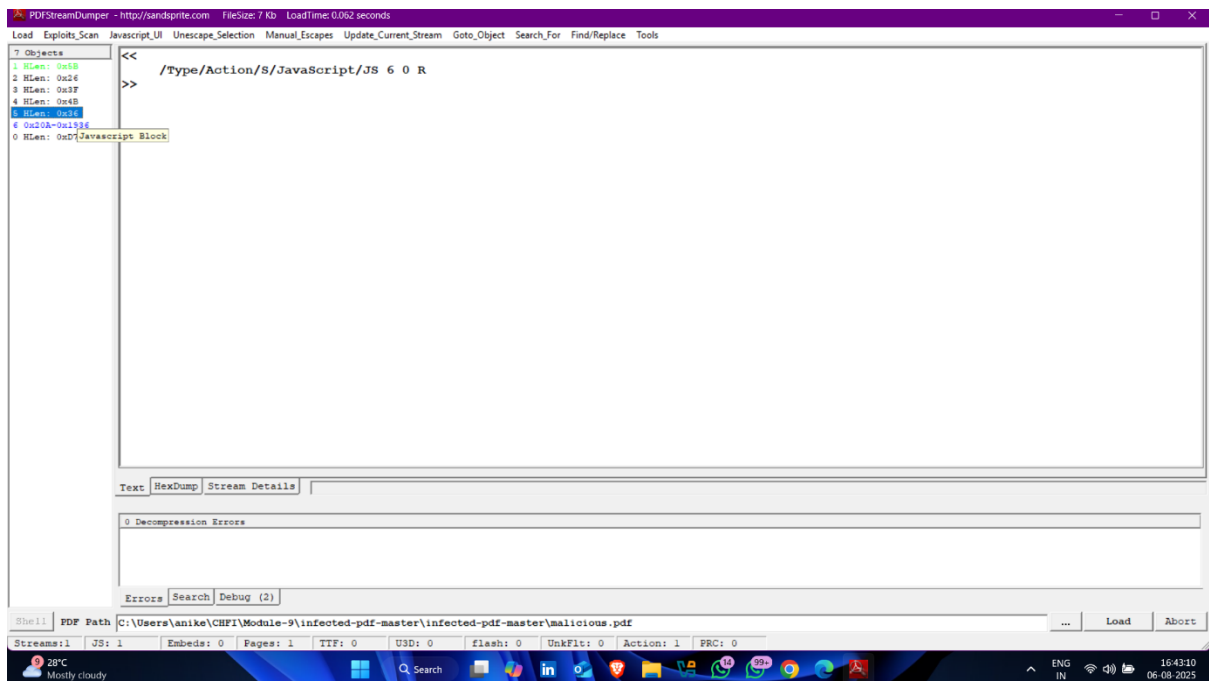
- Then click on open



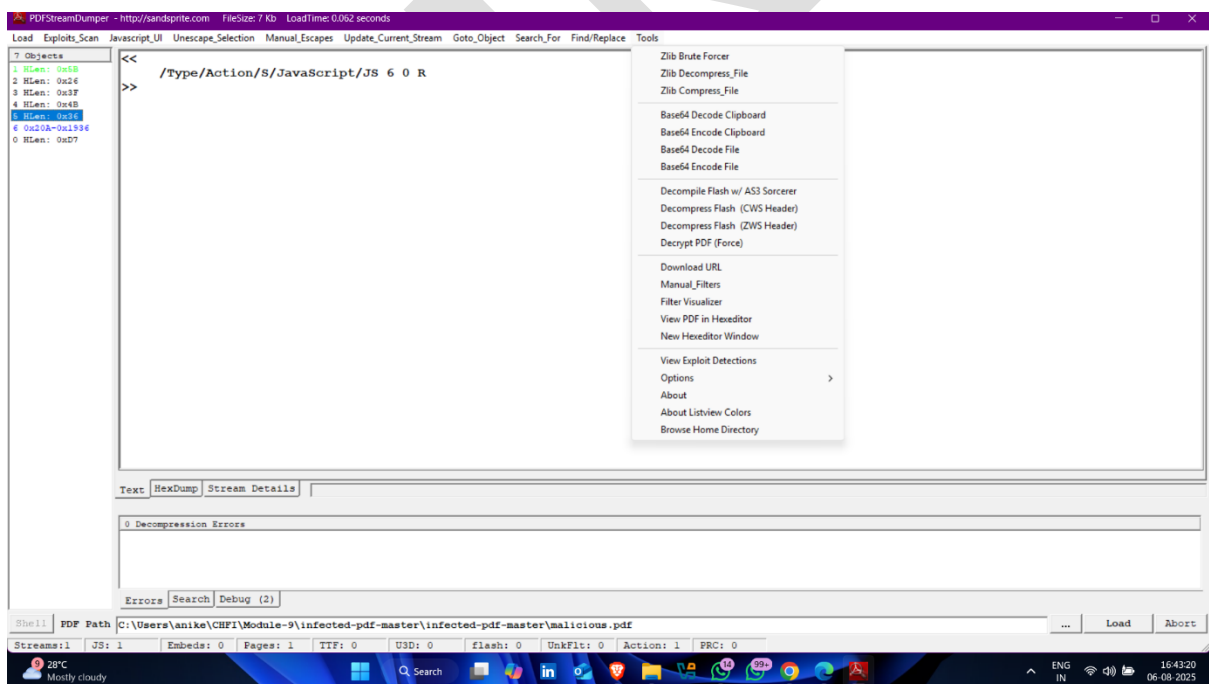
- Pdf uploaded 🖱️



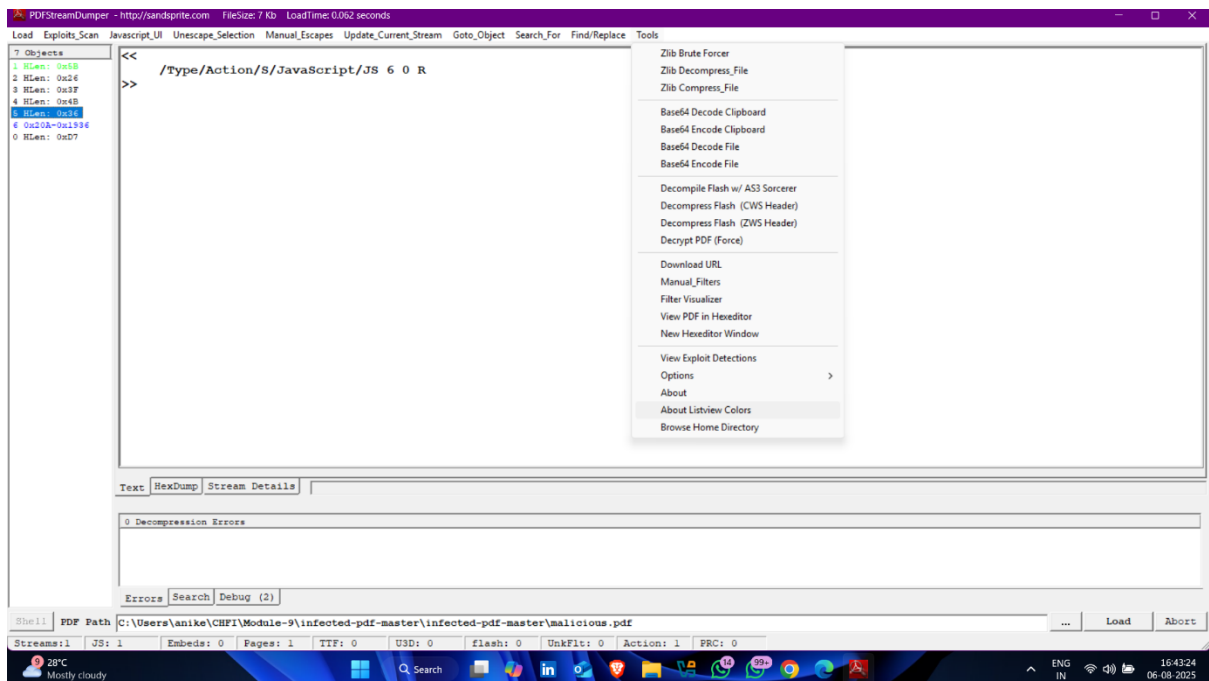
- Now there is many colour headers



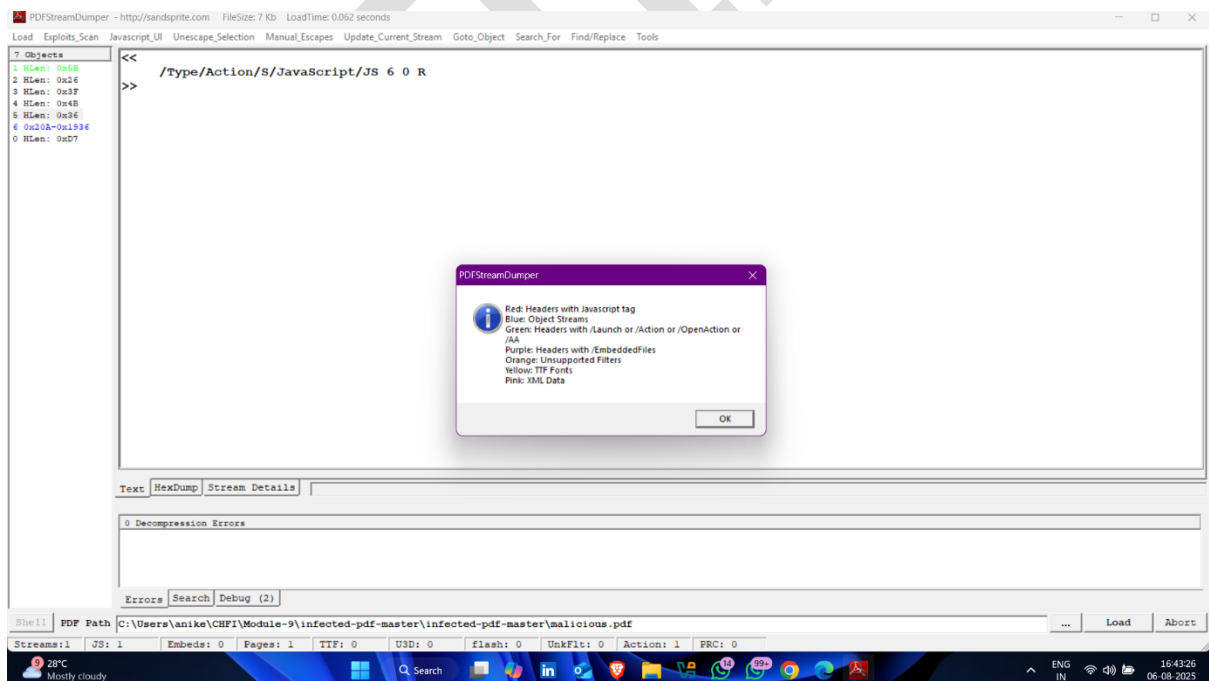
- Click on Tools Section



- Then click on **about listview colors**



- Here, detailed information about each color header – red colour means hidden javascript code in pdf file



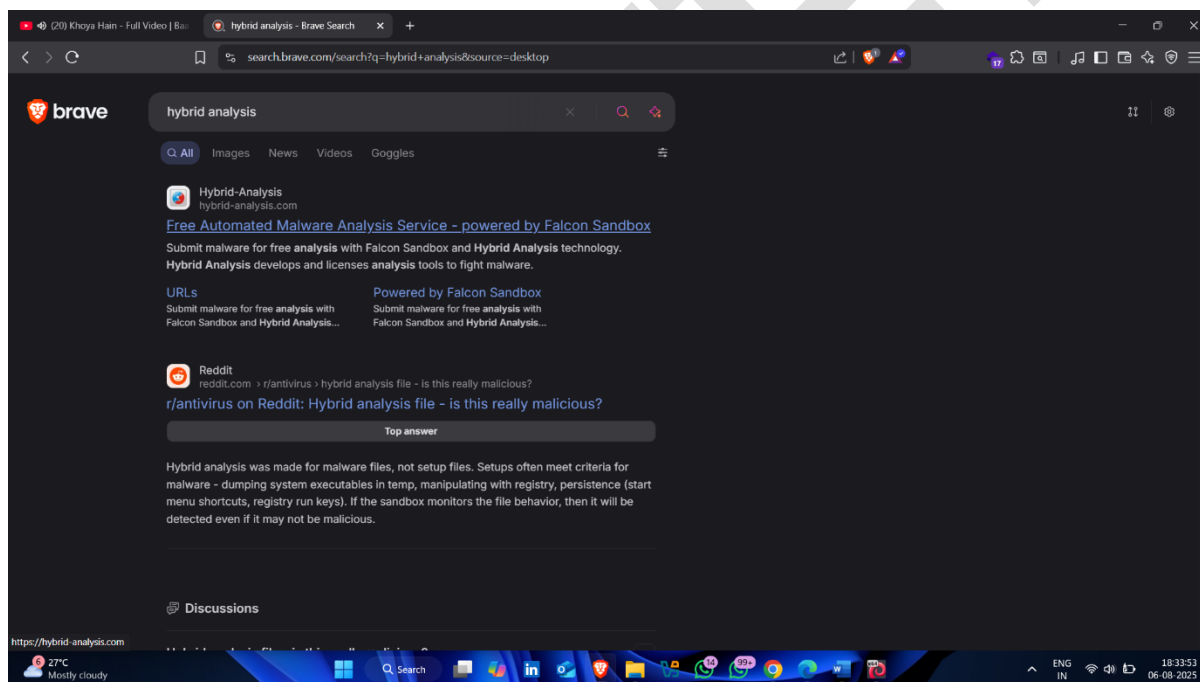
Lab-4 Examine a Suspicious File Using Online Resources

Examine a Suspicious File Using Online Resources" means uploading or checking a file (or its hash) on trusted online analysis platforms to determine if it is **malicious**, **benign**, or **potentially unwanted** — without executing it on your own system.

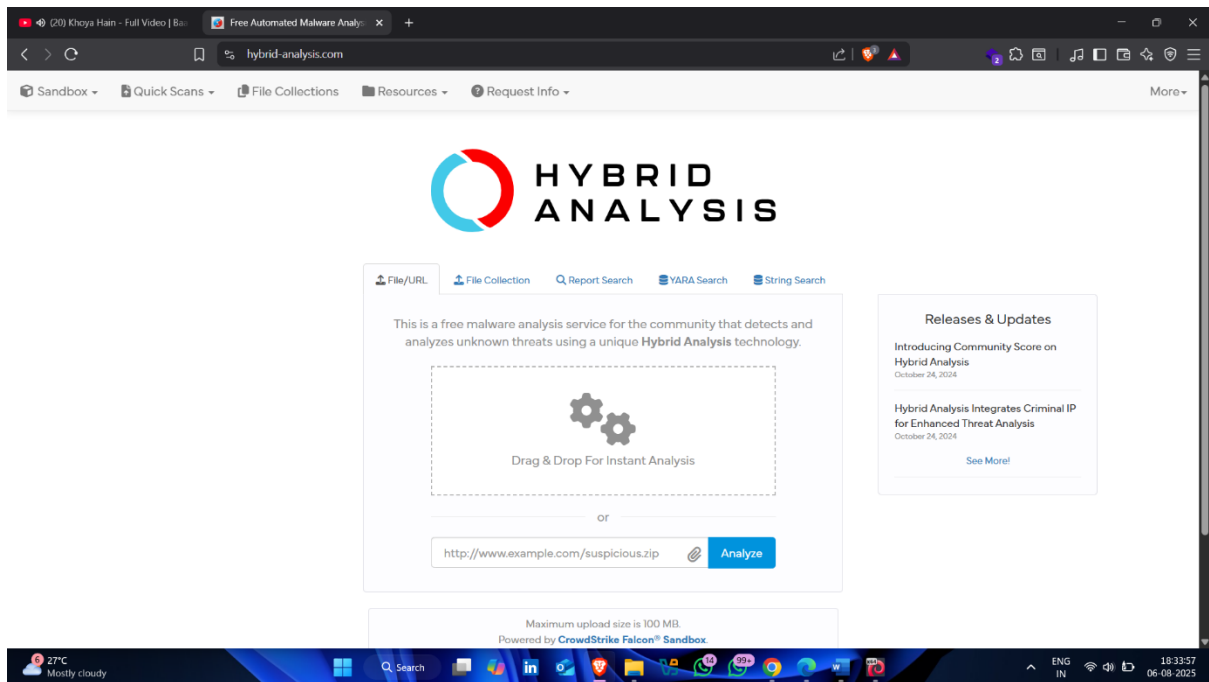
This step is common in **malware analysis** and **digital forensics** to get quick intelligence before doing deep static/dynamic analysis.

How to use it -:

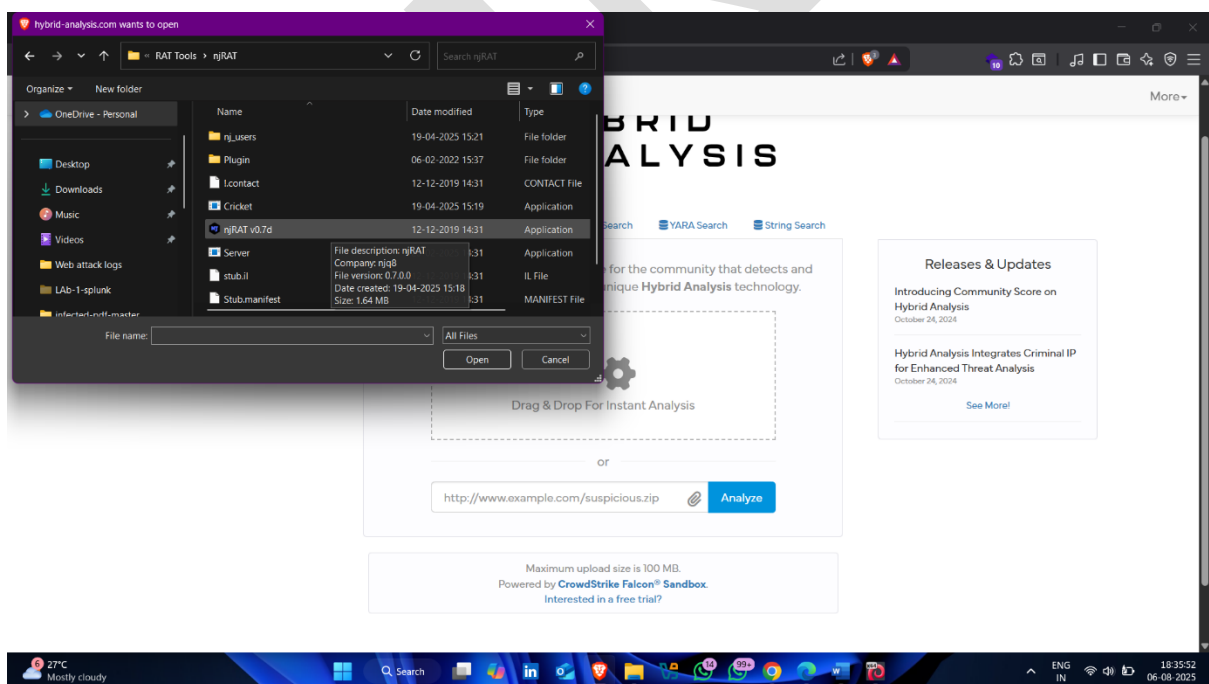
- Open browser and search **hybrid analysis**



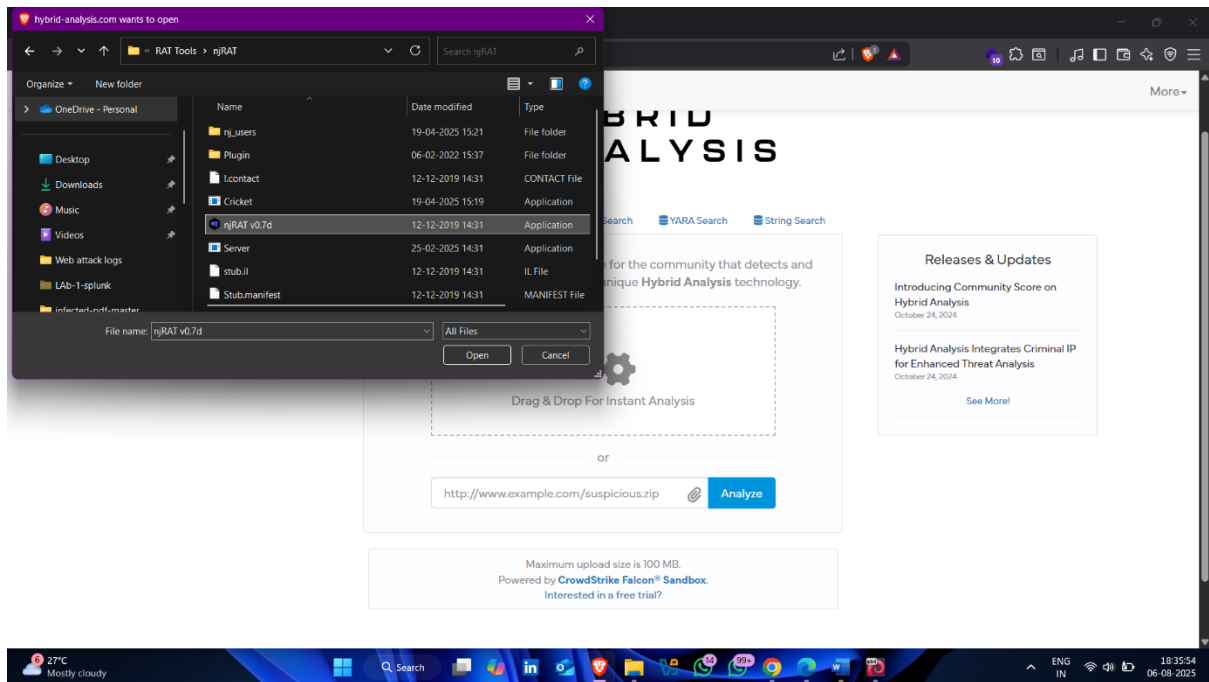
- Click on setting icon



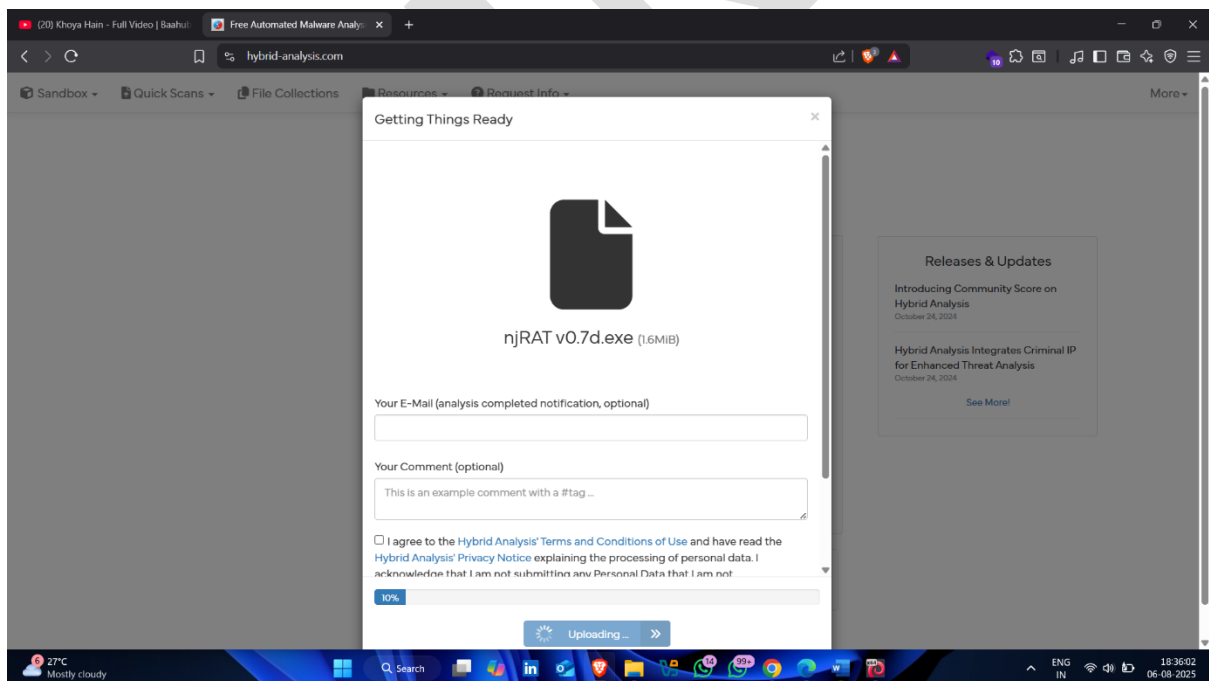
- Now select file that you want to analyze



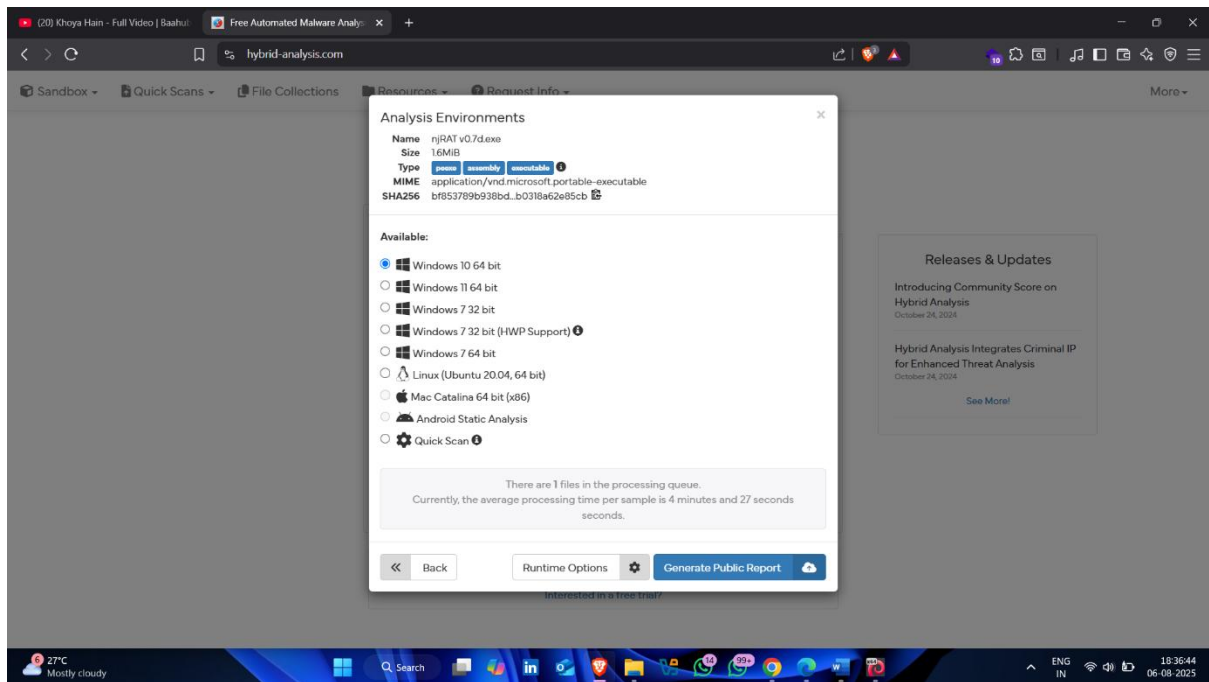
- Click on open



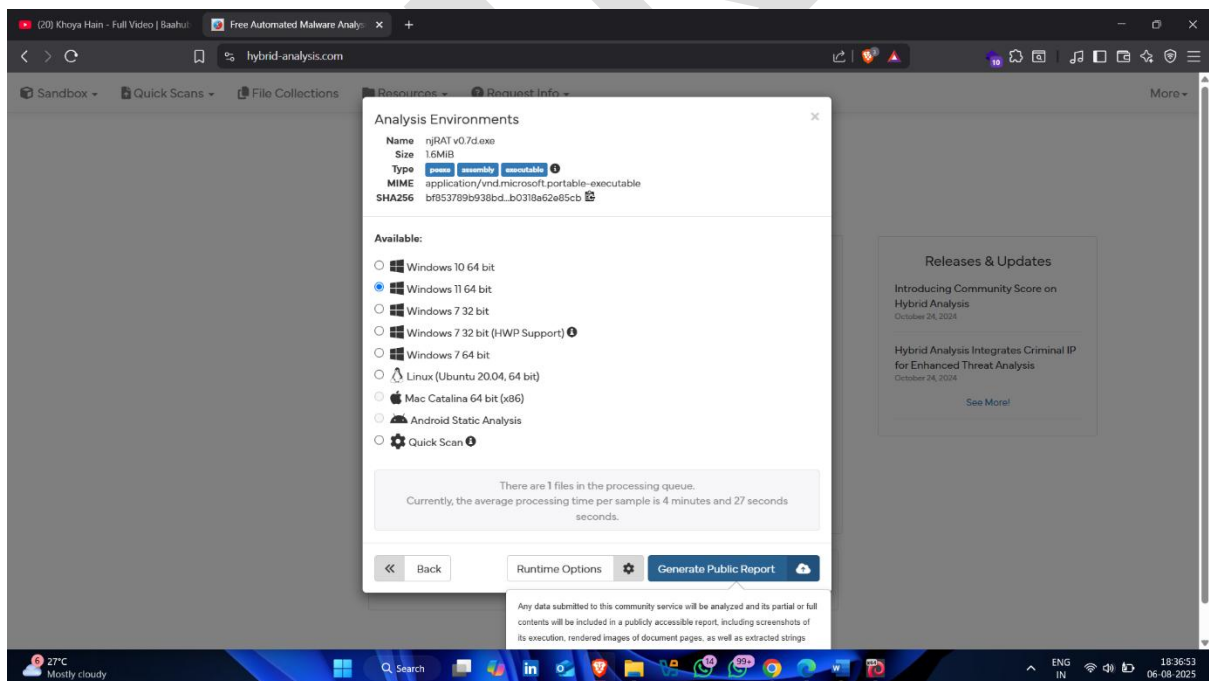
- File uploading started



- Select operating system



- And then click on **Generate public report**



- Here , detailed about file and its malicious

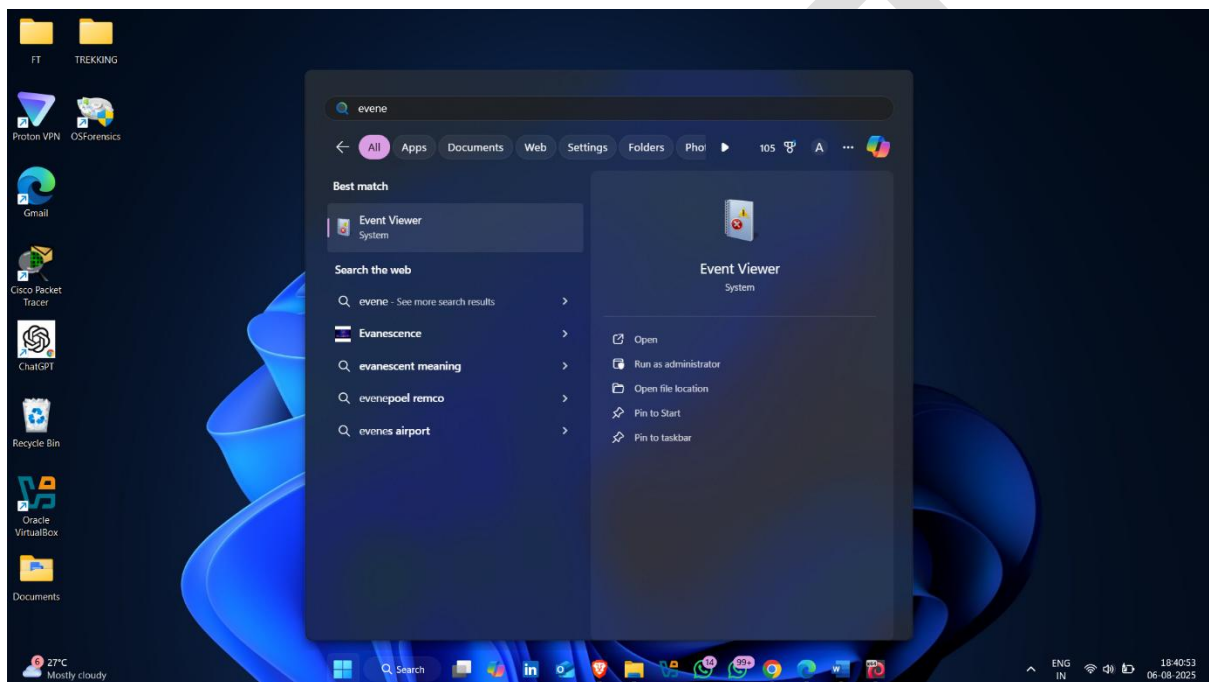
The screenshot shows the Hybrid Analysis website interface. The main heading is "Analysis Overview". On the left, submission details for "njRAT v0.7d.exe" are listed: Size (1.6MiB), Type (executable), Mime (application/x-dosexec), SHA256 (bf853789b938bd5da8aeb52511379a332c7cf238266a21bfc0318a62e85cb), Submitted At (2017-07-05 22:40:22 (UTC)), Last Anti-Virus Scan (2025-01-12 14:40:57 (UTC)), and Last Sandbox Report (2024-03-20 04:50:49 (UTC)). On the right, the "malicious" status is confirmed with a Threat Score of 100/100 and AV Detection of 89%. A list of labels includes #adware, #autorun, #backdoor, #crypt, #downloader, #exploit, #injector, #keylogger, #ransomware, #riskware, #rootkit, #trojan, #worm, and #windows-server-utility. Below this, the "Anti-Virus Results" section shows two scanners: CrowdStrike Falcon (Static Analysis and ML) and MetaDefender (Multi Scan Analysis), both reporting "Malicious (100%)" and "Malicious (18/23)" respectively. A large, diagonal "ANALYSIS" watermark is visible across the bottom half of the image.

Lab-5 Examine Windows Event Log

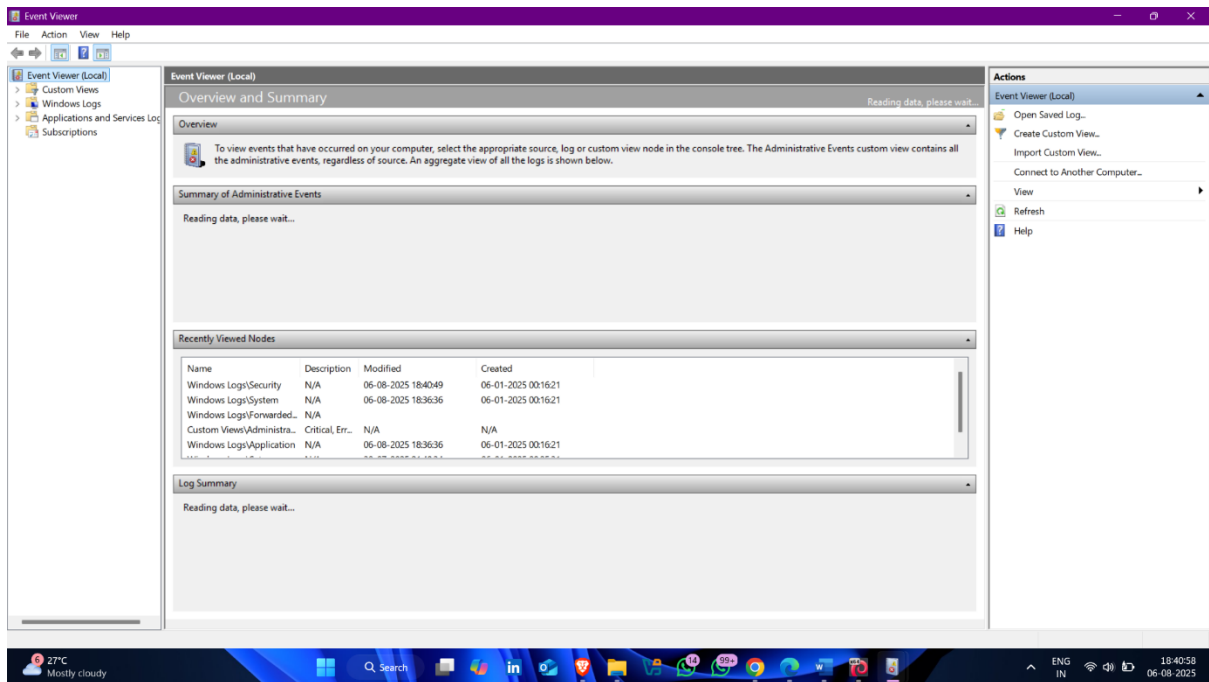
Examine Windows Event Log means reviewing the logs that Windows automatically records about **system, security, and application activities** to detect issues, investigate incidents, or perform forensic analysis.

How to do it -:

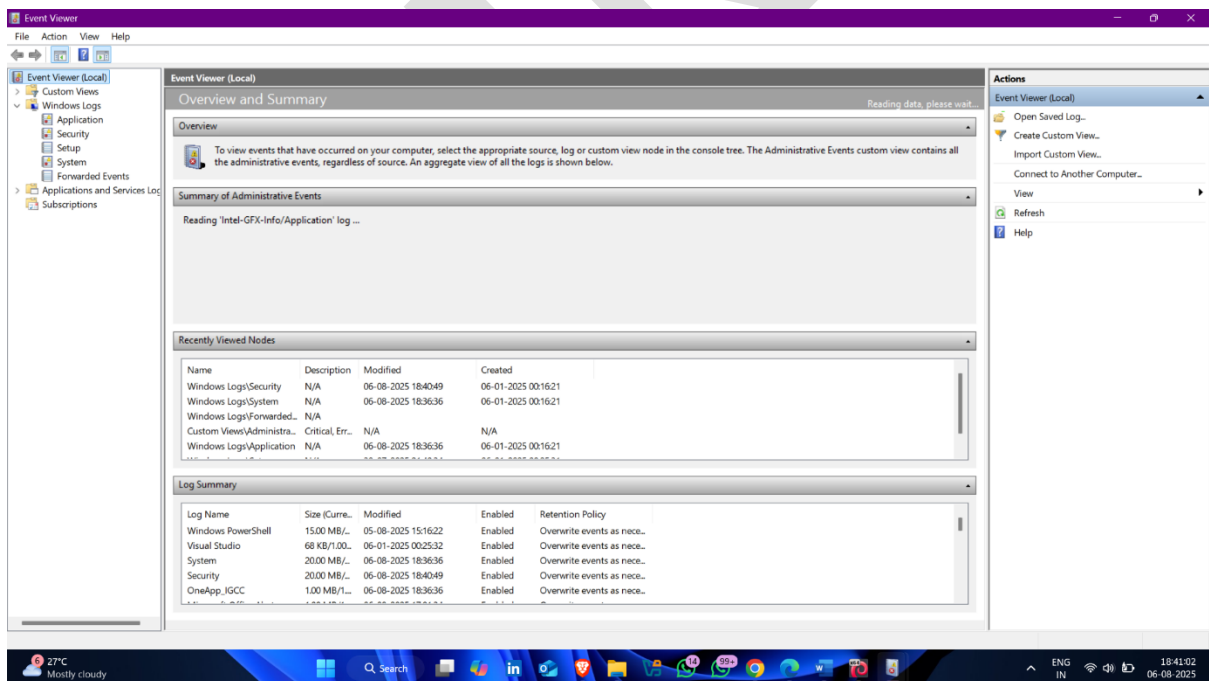
- Open windows **Event Viewer**



- Different logs structure



- Click on **windows logs**



- Click on **application** and **analyze the logs**

